

Transcriptomics reveals the molecular regulation of Chinese medicine formula on improving bone quality in broiler

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ABSTRACT Skeletal disorder is of concern to the poultry industry as it affects animal welfare and production performance. Traditional Chinese medicine could improve bone quality and reduce the incidence of bone disease, but the molecular regulation of Chinese medicine formula (CMF) on improving bone quality in broilers is still unclear. This study was performed to research the effects of CMF on skeletal performance of Cobb broilers and reveal the molecular regulation. A total of 120 one-day-old Cobb broilers were randomly allocated into 4 equal groups of 30 chickens, with 5 replicates and 6 chickens in each replicate. The control (CON) group was fed a diet without CMF, while the CMF1, CMF2, and CMF3 groups were supplemented with different CMF at 6,000 mg/kg diet, respectively. The broilers were raised to 60 d of age, then bone tissues were collected for biomechanical properties, micro-CT detection and transcriptomic sequencing analysis. The results showed that CMF3 improved the biomechanical properties of broiler tibia, via increasing the elastic modulus ($P < 0.05$), yield

strength ($P > 0.05$), maximum stress ($P < 0.05$) and fracture stress ($P < 0.05$) of the tibia. Micro-CT analysis indicated that CMF3 increased the bone mineral density (BMD), bone volume/total volume (BV/TV), bone surface density (BS/TV), trabecular number (Tb.N), trabecular thickness (Tb.Th), and decreased the trabecular separation (Tb.Sp) of femur cancellous bone ($P < 0.05$). RNA-seq analysis revealed 2,177 differentially expressed genes (DEGs) ($|\log_2\text{FoldChange}| \geq 1$, FDR < 0.05) between the CMF3 group and CON group. Kyoto Encyclopedia of Genes and Genomes pathway (KEGG) analysis showed 13 pathways mostly associated with bone growth and development and bone metabolism, and we identified 39 bone-related DEGs. This study suggests that CMF3 could improve bone strength and bone microstructure of broilers, and showed a positive effect on bone performance. Our research could provide a theoretical reference for the development of pollution-free feed additives to improve the skeletal performance of broilers, which could help promote healthy farming of chickens.

Key words: Chinese medicine formula, broiler, bone biomechanical properties, bone microstructure, RNA-seq

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INTRODUCTION

It is predicted that poultry production will increase by 1.5% between 2020 and 2029, and that global poultry meat production will reach about 146 million tons by 2029 (OECD/FAO, 2020). Broiler products account for the majority of poultry meat products. In order to meet the market demand, the genetic breeding of modern broilers is mainly aimed at fast growth and rapid weight gain. Over the past few decades, the growth

rate of broilers has increased by more than 400% and the feed conversion rate has decreased by 50% (Zuidhof et al., 2014). This means that broilers are more productive and take significantly less time to reach high body weight. However, the breeding goal of rapid body weight gain has brought higher economic benefits to the broiler industry, but it has also caused serious bone problems for broilers (Kierończyk et al., 2017). The main reason is that the speed of bone growth and development of broilers cannot keep up with the speed of body weight gain, resulting in higher porosity and lower salinity of bone, which seriously affects bone quality (Williams et al., 2004). On the contrary, broilers with slower growth rate have better bone quality, which showed lower porosity, higher salinity, fracture strength and ash content (Williams et al., 2004; Santos et al., 2022).

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As important physiological characteristics of domestic chickens, bone traits are closely related to the growth, movement and health. Currently, the common bone diseases in modern broiler production are tibial dyschondroplasia, femoral head necrosis, valgus-varus deformity, rickets, tenosynovitis (Joiner et al., 2005; Dinev, 2012; Tang et al., 2020a; Liu et al., 2021; Xu et al., 2023). Impaired leg bone development can increase the susceptibility to fracture of broiler, bone which is closely related to the mobility ability, feeding conditions, and health status of broilers, and leads to poor growth, high mortality, decreased carcass quality, and slaughter rate of broilers (Xu et al., 2022a). Impaired leg development seriously affects animal welfare and economic efficiency. Therefore, in the pursuit of rapid growth and weight gain, it is beneficial to the development and progress of the domestic chicken industry to find solutions to the skeletal problems of domestic chickens.

Traditional Chinese herbal medicine has the advantages of abundant drug sources, natural, low toxicity, no drug resistance, no residue, etc. After reasonable compatibility, they have been used to prevent and treat various bone diseases, and have achieved great success (Lin et al., 2017; Qian et al., 2021; Li et al., 2022a). Especially in the treatment of osteoporosis and fracture in human and model animals, Chinese herbal medicine plays an important role. Chinese herbal medicines were used to treat bone diseases and improve bone quality by regulating the dynamic balance between bone formation and bone resorption. Studies showed that Chinese herbal medicines used to treat osteoporosis usually promote the bone formation of osteoblasts and inhibit the bone resorption process of osteoclasts, thus effectively alleviating the symptoms of osteoporosis by maintaining the balance of bone metabolism (He et al., 2019; Huai et al., 2021). However, few studies have been done on the application of Chinese herbal medicines in domestic chickens.

At present, there is no specific drug for the prevention and treatment of poultry bone diseases, and the pathogenesis of most poultry bone diseases is still not clear.

In this study, *Angelica sinensis*, *Lycium barbarum*, *Chinese yam*, *Salvia miltiorrhiza*, and other 16 kinds of Chinese herbs commonly used to treat bone diseases were selected, and synthesized 3 different Chinese medicine formulas (CMF) according to different proportions, and the purpose of this study was to evaluate the differences in skeletal phenotypes of broilers after feeding different CMF. RNA-Seq was performed on the bone

tissues of broilers in the optimal CMF group and the standard diet group to investigate whether the CMF could affect the expression of genes involved in bone growth and development, and to reveal the potential molecular mechanism of its influence on bone growth and development. This study will provide theoretical basis for developing pollution-free feed additives to improve bone properties of broilers, in order to effectively reduce the incidence of bone disease and promote healthy breeding of domestic chickens.

MATERIALS AND METHODS

Ethics Approval and Consent to Participate

The research was carried out at the Faculty of Animal Science and Technology, Yunnan Agricultural University. The protocol used in this study was approved by the Animal Care and Use Committee of Yunnan Agricultural University (Approval ID: Yau ACU C01).

Preparation of Chinese Medicine Compound

According to previous study, the basic formula of Chinese herbal medicine has the effect of improving bone characteristics in broilers (Liu et al., 2022d). It was composed of *Angelica sinensis*, *Lycium barbarum*, *Chinese yam*, *Salvia miltiorrhiza*, *Schisandra chinensis*, *Rhizoma corydalis*, *Fructus cnidii*, and *Oyster* (5:5:5:1:5:1.6:2:2.4). Formula I was composed of *Pueraria Lobata*, *Prepared Radix Rehmanniae*, *Epimedium*, and *Rhizoma Cyperi* (5:5:2:2). Formula II was composed of *Fructus Psoraleae*, *Rhizoma Drynariae*, *Achyranthes bidentata*, and *Astragalus* (3:5:3:5). All Chinese herbal monomers were crushed and mixed in proportion to prepare Chinese herbal medicine compound. The dosage of CMF was based on our previous research (Liu et al., 2022d). The experiment consisted of 4 groups, the control (CON) group was fed a diet without any Chinese herbs (Table 2), while CMFs were added at the level of 6,000 mg/kg diet in the CMF1 (Basic formula + Formula I), CMF2 (Basic formula + Formula II) and CMF3 (Basic formula + Formula I + Formula II) groups, respectively. Table 1 shows the detailed treatment of traditional Chinese medicine formula.

Animals and Management

One-day-old Cobb broilers were purchased from the Kunming ZhengDa Co., Ltd. (Kunming, China). A total

Table 1. Group of chicken and addition of Chinese medicine.

Chinese medicine/Groups	Control (CON)	Chinese medicine formula 1 (CMF1)	Chinese medicine formula 2 (CMF2)	Chinese medicine formula 3 (CMF3)
Basic formula	—	+	+	+
Formula I	—	+	—	+
Formula II	—	—	+	+

CON group: basal diet; CMF1 group: basal diet + 0.6% CMF1; CMF2 group: basal diet + 0.6% CMF2; CMF3 group: basal diet + 0.6% CMF3. "+" indicates add, "—" indicates do not add.

Table 2. Compositions and nutrient levels of the chicken diets used in the experiment.

Dietary component	Period I ¹ (%)	Period II ¹ (%)
Maize	64.70	60.90
Soy protein	30.16	25.12
Wheat bran	0.00	10.00
Soybean oil	1.10	0.00
CaHPO ₄ ·2H ₂ O	1.50	1.50
Stone meal	0.70	0.60
Middling flour	0.41	0.46
Methionine	0.08	0.07
Salt	0.35	0.35
Minerals and vitamins mix ²	1.00	1.00
Total	100.00	100.00
Nutritional composition ³		
Metabolism energy /(MJ/kg)	12.13	11.97
Crude protein	20.00	19.50
Calcium	0.90	0.80
Available phosphorus	0.37	0.37
Lysine	0.98	0.75
Methionine	0.39	0.32

¹Period I is age 1–42 d; Period II is older than 42 d of age.

²Supplied per kilogram of diet: vitamin A, 15,000 U; vitamin D 3, 3,300 U; vitamin E, 62.5 mg; vitamin K, 3.6 mg; thiamine, 3.0 mg; riboflavin, 9.0 mg; pyridoxine, 6 mg; cyanocobalamin, 0.03 mg; niacin, 60 mg; calcium pantothenate, 18 mg; folic acid, 1.5 mg; biotin, 0.36 mg; choline chloride, 600 mg; Fe, 80 mg; Cu, 12 mg; Zn, 75 mg; Mn, 60 mg; I, 0.35 mg; Se, 0.15 mg; and anticoccidial (chloropyridine 125 mg/kg).

³Nutrient levels were calculated values.

of 120 Cobb broilers were raised in the teaching practice chicken farm of Yunnan Agricultural University (Kunming, China). A total of 120 one-day-old Cobb broilers were randomly allocated into 4 equal groups of 30 chickens, with 5 replicates and 6 chickens in each replicate. The chickens were reared under standard conditions on starter diets (Period I: 20.0% crude protein, 12.13 MJ/kg metabolizable energy) from 0 to 42 d of age and then on regular diet (Period II: 19.5% crude protein, 11.97 MJ/kg metabolizable energy) to 60 d. The diet formula conforms to Chinese Broiler Feeding Standard (NY/T33-2004, Ministry of Agriculture of the People's Republic of China, 2004) recommendations. Dietary composition is shown in Table 2. Chickens were given free access to food and water throughout the rearing period. Both males and females were raised in the same cage (90 × 70 cm) with a stocking density of 15 birds/m² for 0- to 14-days old, 6 birds/m² for 15- to 28-days old, and 3 birds/m² for 29- to 60-days old. A total of 15 cages per treatment group, assigned according to the age and replicate of the chickens. The whole experiment period was carried out according to the feeding management of Yunnan Agricultural University, that is, 1- to 7-days old, temperature 33°C to 35°C, relative humidity 65 to 70%, light 24 h; 8- to 14-days old, temperature 30°C to 33°C, relative humidity 60 to 65%, light 23 h; 15- to 21-days old, temperature 28°C to 30°C, relative humidity 55 to 60%, light 18 h; 22- to 28-days old, temperature 26°C to 28°C, relative humidity 55 to 60%, light 18 h; 29- to 35-days old, temperature 24°C to 26°C, relative humidity about 55%, light 18 h; 36- to 42-days old, temperature 21°C to 24°C, relative humidity about 55%, light 18 h; 43- to 60-days old, temperature 18°C to 21°C, relative humidity about 55%, light 18 h.

Determination of Growth Performance

During the rearing of the test chickens, the live weight of each chicken was measured every 14 d in the morning. The amount of feed and the amount of splattered feed outside the tank are recorded daily to calculate feed intake. The live weight, average daily gain (ADG), average daily feed intake (ADFI), and feed conversion ratio (FCR) were calculated at 28 and 56 d, respectively.

ADG (g): ADG is the difference between the final weight of each stage (28 d) minus the initial weight of the stage divided by the number of feeding days.

ADFI (g): ADFI is equal to the total feed intake of each stage (28 d) divided by the feeding days of this stage and then divided by the number of chickens tested.

FCR: FCR is equal to ADFI divided by ADG.

Sample Collection

No feed and no drinking water were allowed for 16 h and 12 h before slaughter. At 60 d of age, 4 broilers were randomly selected and slaughtered per replicate, that is, 20 broilers per treatment group. Chickens were slaughtered for cervical dislocation according to national standards for slaughter of laboratory animals. The intact tibia and femur of the left leg were collected, respectively, and other tissues attached to the surface were removed. Then, they were washed with 0.75% normal saline, kept moist and sealed in time, and finally stored at −20°C for determination of bone biomechanics and bone microstructure. The right subchondral cancellous bone tissue samples of tibia were collected and divided into multiple 2 mL sterile centrifuge tubes (RNAase-free). The samples were quickly frozen in liquid nitrogen and then stored in an ultra-low temperature refrigerator at −80°C for subsequent transcriptome analysis and real-time PCR. The chicken carcasses after sample collection were treated harmlessly in accordance with the Guidelines for the Care and Use of Experimental Animals of Yunnan Agricultural University.

Biomechanical Examination of Bone

Bone biomechanical examination was carried out with an AG-IS electronic universal testing machine (Shimadzu Corporation, Kyoto, Japan). After thawing, the 10 left tibia of each group ($n = 10$) was used as the test sample, and the bone biomechanical indexes such as elastic modulus, yield strength, maximum stress, and fracture stress (N/mm²) were detected by 3-point bending test (Harash et al., 2020). The sample was placed on the 2 support points of the electronic universal testing machine, and a downward load was applied to the specimen at the midpoint of the 2 support points. Three-point bending occurred when the 3 contact points of the sample formed 2 equal moments. The loading rate was 2 mm/min and the loading was uniform until the sample

was destroyed. The formula for calculating the distance of the supporting rail is: $L = (a + 3b) \pm 0.5b$, where a (mm) is the diameter of the supporting rail and b (mm) is the diameter of the tibia.

Examination of Bone Microstructure

The 10 intact left femur of each group ($n = 10$) was placed into the small animal CT imaging system (Micro-CT, ZkS-McTsharp; the data processing software was ZKS-micro-CT 4.1), and the scan was conducted from the top 3 mm of the femoral condyle (distal) to the middle part (proximal) of the femur (Distinguish the distal and proximal ends of the femur by using the mid-diaphysis of the femur as the starting point, with the mid-diaphysis as the proximal end and the top of the malleolar bone as the distal end.). The scanned sections were used to analyze the microstructure of the cancellous bone in the distal femur. Parameters related to trabecular characteristics of cancellous bone of distal femur: bone mineral density (BMD, g/cm³), bone volume/total volume (BV/TV, %), bone surface density (BS/TV, 1/mm), trabecular number (Tb.N, 1/mm), trabecular thickness (Tb.Th, μ m), trabecular separation (Tb.Sp, μ m).

Transcriptomic Analysis

RNA Extraction, cDNA Library Construction, and RNA-Seq On 60 d, the subchondral cancellous bone tissue of tibia from 1 chicken per replicate in the CMF3 ($n = 5$) and CON ($n = 5$) groups was collected for RNA-Seq. Sample preparation and RNA-Seq were performed as described by Dou et al. (2022). Total RNA was extracted from 2 groups of chicken bone tissue samples using Takara MiniREST general RNA extraction kit (TaKaRa Biotechnology Co., Ltd., Dalian, PR China) according to the manufacturer's protocol. RNA purity was determined by NanoDrop 2000 spectrophotometer (Nanodrop, Wilmington, DE). Total RNA concentration was determined by Qubit 2.0 fluorimeter (Thermo Fisher Scientific Inc., Waltham, MA) and Agilent 2100 Bioanalyzer (Agilent TECHNOLOGIES, Inc., Santa Clara, CA) detects integrity. mRNA was isolated from total RNA samples using oligo (dT) magnetic beads (Invitrogen, Carlsbad, CA). RNA-seq sample preparation kits (Illumina, San Diego, CA) were used to construct cDNA libraries with the generated mRNA. Then the sequencing was performed on Illumina sequencing platform (Illumina HiSeq2500).

Quantification and Analysis of Raw Data To ensure the quality of the samples, the raw sequence data were examined using software FastQC (Version 0.11.5, <https://github.com/s-andrews/FastQC>). Then the raw data were filtered with software Trim Galore (Version 0.4.4, <https://github.com/FelixKrueger/TrimGalore>) to obtain clean data. Clean data were compared with the chicken reference genome (ftp://ftp.ensembl.org/pub/release-95/fasta/gallus_gallus/dna/) by software HISAT2 (Version 2.0.4, [\[hisat2\]\(https://github.com/infphilo/hisat2\)\) to obtain SAM files. The SAM files were converted to BAM files using software Samtools \(Version 1.5, <https://github.com/samtools/samtools>\). The BAM files were compared with the *Gallus gallus* genome annotation file \(\[ftp.ensembl.org/pub/release-95/gtf/gallus_gallus/\]\(ftp.ensembl.org/pub/release-95/gtf/gallus_gallus/\)\) using software HTSeq \(Version 0.9.0, <https://github.com/simon-anders/htseq>\) to complete the gene annotation and obtain the gene counts of each sample in different groups.](https://github.com/infphilo/</p>
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Differentially Expressed Genes Analysis The software DESeq2 (Version 1.20.0, <https://github.com/mikelove/DESeq2>), R language (Version 3.5.1, <https://www.r-project.org/>) and its auxiliary software platform RStudio (Version 1.1.463, <https://www.rstudio.com/>) for differentially expressed gene (DEG) analysis. PADJ (FDR, Benjamini-Hochberg) represents the significance of DEGs, and log₂FoldChange (Log₂FC) represents the relative gene expression level. We defined Log₂FC ≥ 1 and FDR < 0.05 to finally obtain DEGs (Dou et al., 2022).

Gene Ontology and Kyoto Encyclopedia of Genes and Genomes Pathway Enrichment Analysis Gene ontology (GO) enrichment analysis of DEGs was conducted combined with gplots (Version 3.0.1.0, <https://CRAN.R-project.org/package=gplots>) and ClusterProfiler (Version 3.8.1, <https://bioconductor.org/packages/clusterProfiler/>). The Kyoto Encyclopedia of Genes and Genomes (KEGG, <http://www.genome.jp/kegg/kegg2.html>) database was used to analyze the pathways of DEGs and to investigate their possible biological functions.

Visual Analysis of Protein-Protein Interaction Network Protein-protein interaction (PPI) network analysis was performed in the String (<https://string-db.org/cgi/input.pl>) database, and then the software Cytoscape (Version 3.9.1, <https://cytoscape.org/>) for visualization of the PPI network.

Quantitative Real-Time PCR Thirty-nine DEGs in KEGG pathways associated with bone growth and development were selected for quantitative real-time PCR (qRT-PCR). Total RNA was extracted from the subchondral cancellous bone tissue samples ($n = 5$, per group) of the same test chickens as the RNA-Seq, and cDNA was synthesized using the reverse transcription kit produced by Takara Corporation (TaKaRa Biotechnology Co., Ltd., Dalian, PR China). SYBR premix Ex TaqTM II kit (TaKaRa Biotechnology Co., Ltd., Dalian, PR China) and the Bio-Rad CFX96 real-time quantitative polymerase chain reaction platform (Bio-Rad Laboratories, Inc., Hercules, CA) were used to perform the qRT-PCR. The mRNA relative expression was analyzed by the 2^{- $\Delta\Delta$ CT} method, and the results were normalized using β -actin as the reference gene. The primers (Qingke Xinye Biotechnology Co., Ltd., Beijing, China) for DEGs are shown in Additional file 1: Table S1.

Statistical Analysis

Excel 2021 was used for data collation, and Software GraphPad Prism 8.0 (GraphPad Software Inc., San

Diego, CA) was used for data analysis and mapping. The data were analyzed by independent samples *T* test for significance of differences for 2 groups, and by 1-way ANOVA for 3 or more groups. All the results were expressed as “mean $\pm$ SEM,” and $P < 0.05$ was considered to be statistically significant.

RESULTS

Comparative Analysis of Growth Performance

The effect of different CMF on live weight, ADG, ADFI and FCR of Cobb broilers is shown in Figure 1. During the whole growing period, the live weight of chickens in each group showed an increasing trend with the increase of age. Compared with the control group, the live weight of CMF1 and CMF3 groups was higher, but there was no statistical difference among all groups ($P > 0.05$). The ADG, ADFI, and FCR of Cobb broilers were no significant differences among all groups ($P > 0.05$).

Comparative Analysis of Bone Biomechanical Indexes

The elastic modulus, yield strength, maximum stress and fracture stress of tibia of broilers were increased by adding Chinese medicine formula in the diet (Figure 2). Specifically, the tibial elastic modulus in CMF1, CMF2, and CMF3 groups was significantly higher than that in CON group ($P < 0.05$, Figure 2A). The tibial yield strength of CMF1 group was significantly higher than

that of CON group ($P < 0.05$); and tibia yield strength in CMF2 and CMF3 groups was higher than that in CON group, but the difference was not statistically significant ($P > 0.05$, Figure 2B). The maximum tibial stress in CMF3 group was significantly higher than that in CON group ($P < 0.05$), the tibial maximum stress in CMF1 and CMF2 groups was higher than that in CON group, but the difference was not statistically significant ($P > 0.05$, Figure 2C). The fracture stress of tibia in CMF2 and CMF3 groups was significantly higher than that in CON group ($P < 0.05$); and tibial fracture stress in CMF1 group was higher than that in CON group, but the difference was not statistically significant ($P > 0.05$, Figure 2D). The elastic modulus and maximum stress of the CMF3 group are the highest, the yield strength of the CMF1 group is the highest, and the fracture stress of the CMF2 group is the highest.

Comparative Analysis of Microstructure of Bone Tissue

The comparison of microstructure of the distal cancellous bone in the femur is shown in Figure 3. Representative 3D Micro-CT images of the distal cancellous bone of the femur in different groups showed that the trabecular bone in the CON group formed a rod-like structure, the trabecular bone became thinner, perforated or even fractured, and the complete trabecular bone plate disappeared, leading to the widening of the remaining trabecular space and the disappearance of the connection between the trabecular bone. The microstructure of femoral cancellous bone in CMF1, CMF2, and CMF3 groups was improved to a certain extent, the distal

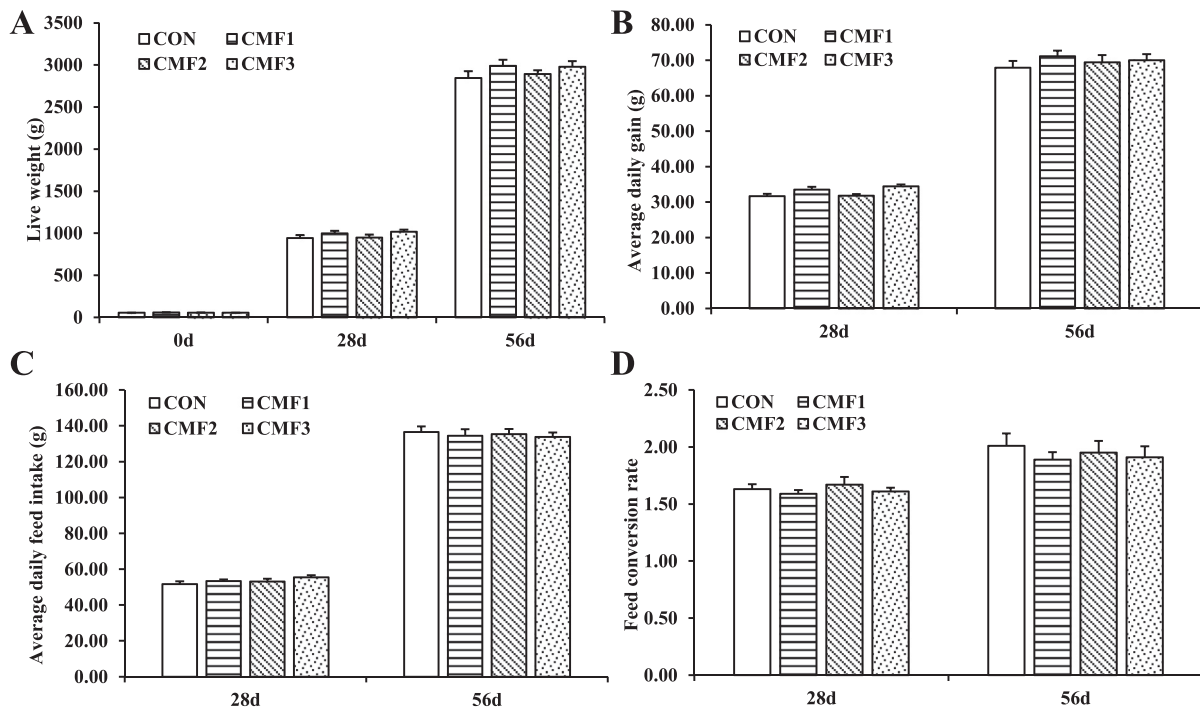


Figure 1. Comparison of growth performance of broilers. (A) Comparison of live weights of broilers. (B) Comparison of average daily gain (ADG). (C) Comparison of average daily feed intake (ADFI). (D) Comparison of feed conversion rate (FCR).

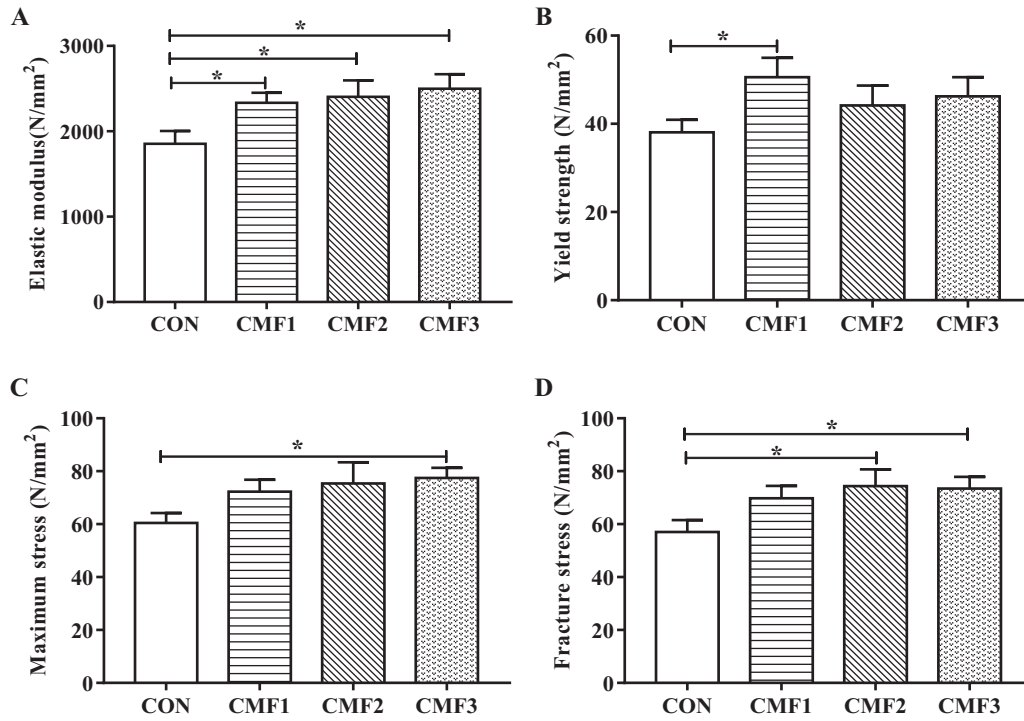


Figure 2. Comparative analysis of tibia biomechanical indexes in broilers ($n = 10$). (A) Comparison of tibial elastic modulus. (B) Comparison of tibial yield strength. (C) Comparison of tibial maximum stress. (D) Comparison of tibial fracture stress. * indicates significant difference ($P < 0.05$).

femoral trabecular bone was partly rod-like and partly plate-like, and the trabecular space was narrowed and the intertrabecular connection increased. In conclusion, the CMF could improve the microstructure of distal femoral cancellous bone of broilers.

The comparison of microstructure parameters of cancellous bone in the distal femur is shown in Figure 4. The specific research results are as follows: The addition of Chinese medicine formula in the diet could improve

the BMD, BV/TV and BS/TV of distal femoral cancellous bone of broilers (Figure 4A–C). Specifically, BMD, BV/TV and BS/TV in CMF groups were higher than those in CON group, with BMD and BV/TV in CMF3 group were significantly higher than those in CON group ($P < 0.05$, Figure 4A and B). BS/TV in CMF3 group was significantly higher than that in CON, CMF1, and CMF2 groups ($P < 0.05$, Figure 4C). The addition of Chinese medicine formula in the diet could increase the

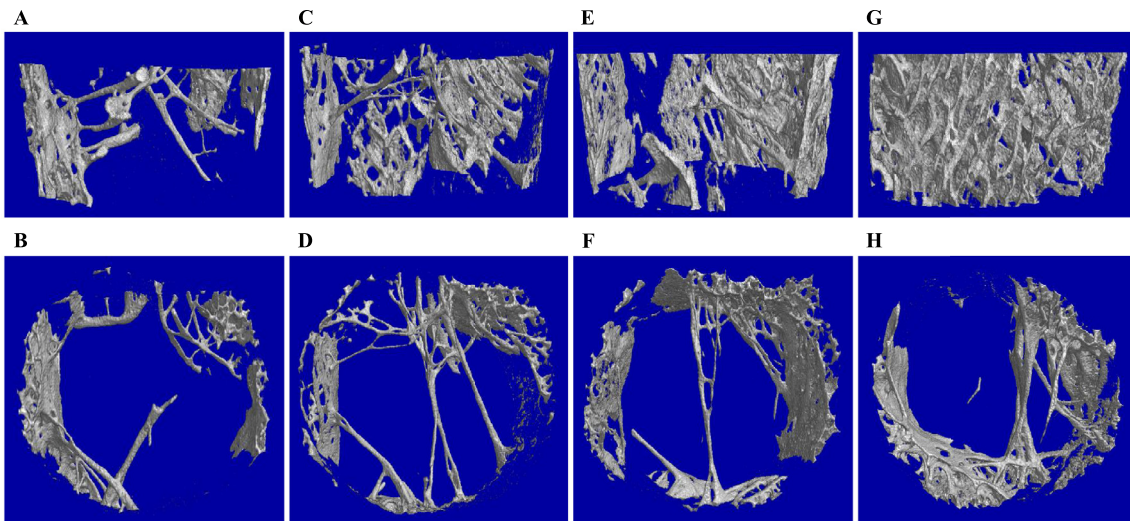


Figure 3. 3D structure of cancellous bone in distal femur of broilers ($n = 10$). (A) Longitudinal section of the distal femoral cancellous bone in CON group. (B) Cross section of the distal femoral cancellous bone in CON group. (C) Longitudinal section of the distal femoral cancellous bone in CMF1 group. (D) Cross section of the distal femoral cancellous bone in CMF1 group. (E) Longitudinal section of the distal femoral cancellous bone in CMF2 group. (F) Cross section of the distal femoral cancellous bone in CMF2 group. (G) Longitudinal section of the distal femoral cancellous bone in CMF3 group. (H) Cross section of the distal femoral cancellous bone in CMF3 group.

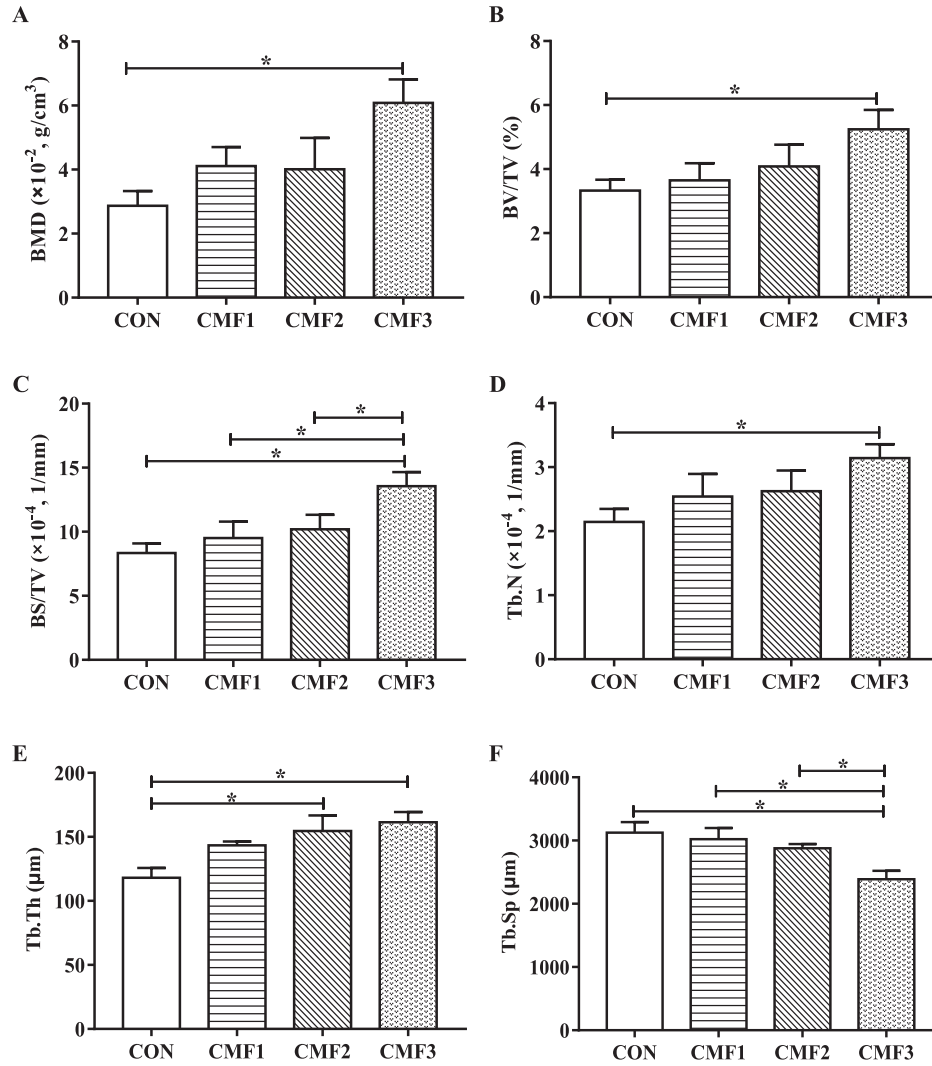


Figure 4. Comparison of microstructural parameters of cancellous bone in the distal femur of broilers ($n = 10$). (A) BMD of cancellous bone in the distal femur. (B) BV/TV of cancellous bone in the distal femur. (C) BS/TV of cancellous bone in the distal femur. (D) Tb.N of cancellous bone in the distal femur. (E) Tb.Th of cancellous bone in the distal femur. (F) Tb.Sp of cancellous bone in the distal femur.

Tb.N and Tb.Th of cancellous bone in the distal femur of broilers (Figure 4D and E). The Tb.N of CMF3 group was significantly higher than that of CON group (Figure 4D), and the Tb.Th of both the CMF2 and CMF3 groups was significantly higher than that of the CON group (Figure 4E). The addition of Chinese medicine formula in the diet could reduce the Tb.Sp of the distal femoral cancellous bone of broilers, and the Tb.Sp in the CMF3 group was significantly lower than that in the CON, CMF1, and CMF2 groups ($P < 0.05$, Figure 4F).

Transcriptome Analysis of Bone Tissue in Broilers

RNA Sequencing and Differential Expression Analysis The sequencing obtained 290.49 Gb raw data, containing 2079.43 million sequences. After filtering the raw data, 285.86 Gb of clean data was retained, which contained 2029.63 million sequences for subsequent

analysis. The sequencing data volume of each sample was above 7G, and the Mapped rate of each sample was higher than 84% (Table 3).

Principal component analysis (PCA) showed that there were significant differences between the CMF3 group and the CON group, and the sample differences between the groups were small (Figure 5B).

A total of 2,177 DEGs were obtained from CMF3 vs. CON, including 1,450 upregulated genes and 727 down-regulated genes (Figure 5A, Additional file 2). Cluster analysis of DEGs showed that the gene expression patterns of DEGs were clustered within the groups, while the differences between the groups were significant (Figure 5C).

Functional Annotation and KEGG Enrichment Analysis of DEGs In order to identify the biological function of DEGs, GO functional enrichment analysis was performed from 2 parts: biological process and cellular component (Figure 6). In biological process, Enriched system development, nervous system development, neurogenesis, cellular developmental process, generation of

Table 3. RNA sequencing data statistics and comparison analysis.

Groups	Sample	Raw data		Clean data		Clean data rate (%)		Mapped rate (%)
		Sequence	Base (bp)	Sequence	Base (bp)	Sequence	Base	
CON	CON1	47,609,279	7,141,391,850	47247341	7,041,459,155	99.24	98.60	88.67
	CON2	46,519,564	6,977,934,600	46119347	6,871,013,773	99.14	98.47	85.79
	CON3	45,600,686	6,840,102,900	45246696	6,750,714,022	99.22	98.69	89.63
	CON4	48,404,629	7,260,694,350	48017329	7,161,205,108	99.20	98.63	84.64
	CON5	53,083,298	7,962,494,700	52587847	7,845,029,046	99.07	98.52	90.1
CMF3	CMF3-1	26,396,294	3,959,444,100	26167206	3,870,333,026	99.13	97.75	89.19
	CMF3-2	29,474,512	4,421,176,800	29255797	4,344,911,615	99.26	98.28	89.74
	CMF3-3	26,487,805	3,973,170,750	26276919	3,902,442,168	99.20	98.22	89.8
	CMF3-4	27,514,801	4,127,220,150	27266146	4,037,109,328	99.10	97.82	89.27
	CMF3-5	34,450,586	5,167,587,900	34228471	5,086,433,103	99.36	98.43	91.23
Total		2079.43 million	290.49 GB	2029.63 million	285.86 GB			

Raw data: raw data volume statistics, sequencing reads, and total number of bases for each sample; Clean data: statistics of valid data, reads, and total number of bases after quality pretreatment for each sample; Clean data ratio: Effective data ratio, clean reads/raw reads*100.

neurons, and other 58 GO terms (Figure 6A and B). In the cellular component, Enriched extracellular region part, extracellular matrix, proteinaceous extracellular matrix, plasma membrane, cell periphery, and other 26 GO terms (Figure 6C and D).

Analysis of KEGG pathways in Cobb broilers showed that 13 KEGG pathways were significantly enriched (Table 4). Including melanogenesis, Wnt signaling

pathway, ECM-receptor interaction, mTOR signaling pathway, TGF-beta signaling pathway, Adrenergic signaling in cardiomyocytes, AGE-RAGE signaling pathway in diabetic complications, MAPK signaling pathway, focal adhesion, adherens junction, neuroactive ligand-receptor interaction, apelin signaling pathway and cell adhesion molecules (CAMs). Among them, the DEGs enriched in these pathways are shown in Figure 7.

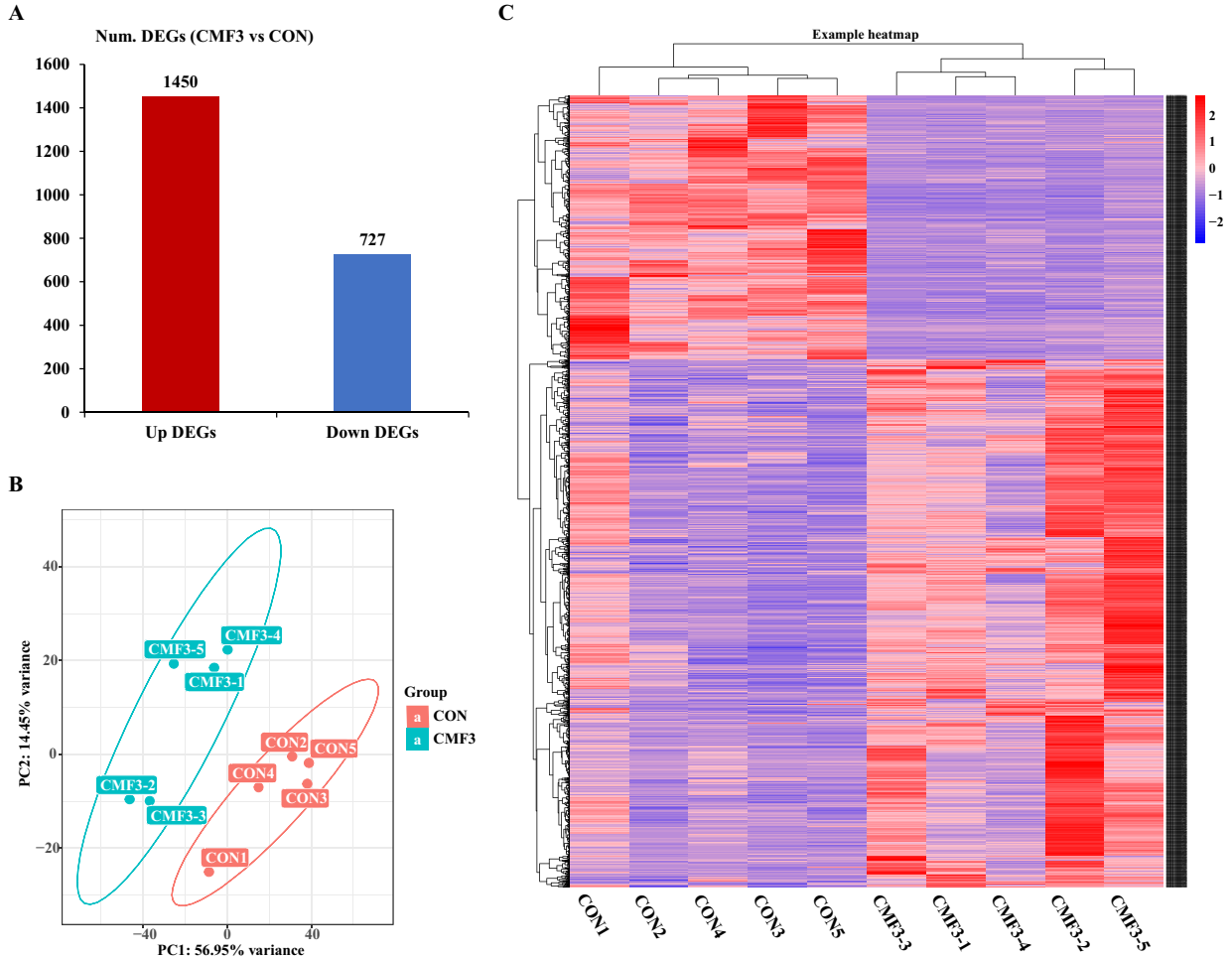


Figure 5. Principal component analysis (PCA) diagram and heatmap of differentially expressed genes (DEGs) ($n = 5$). (A) The number of DEGs (up/downregulated) after CMF3 treatment. (B) PCA diagrams from CMF3 (blue) and CON (red) groups. (C) Heatmap of 2,177 DEGs from the CMF3 vs. CON groups, of which 1,450 DEGs were upregulated (red) and 727 DEGs were downregulated (blue).

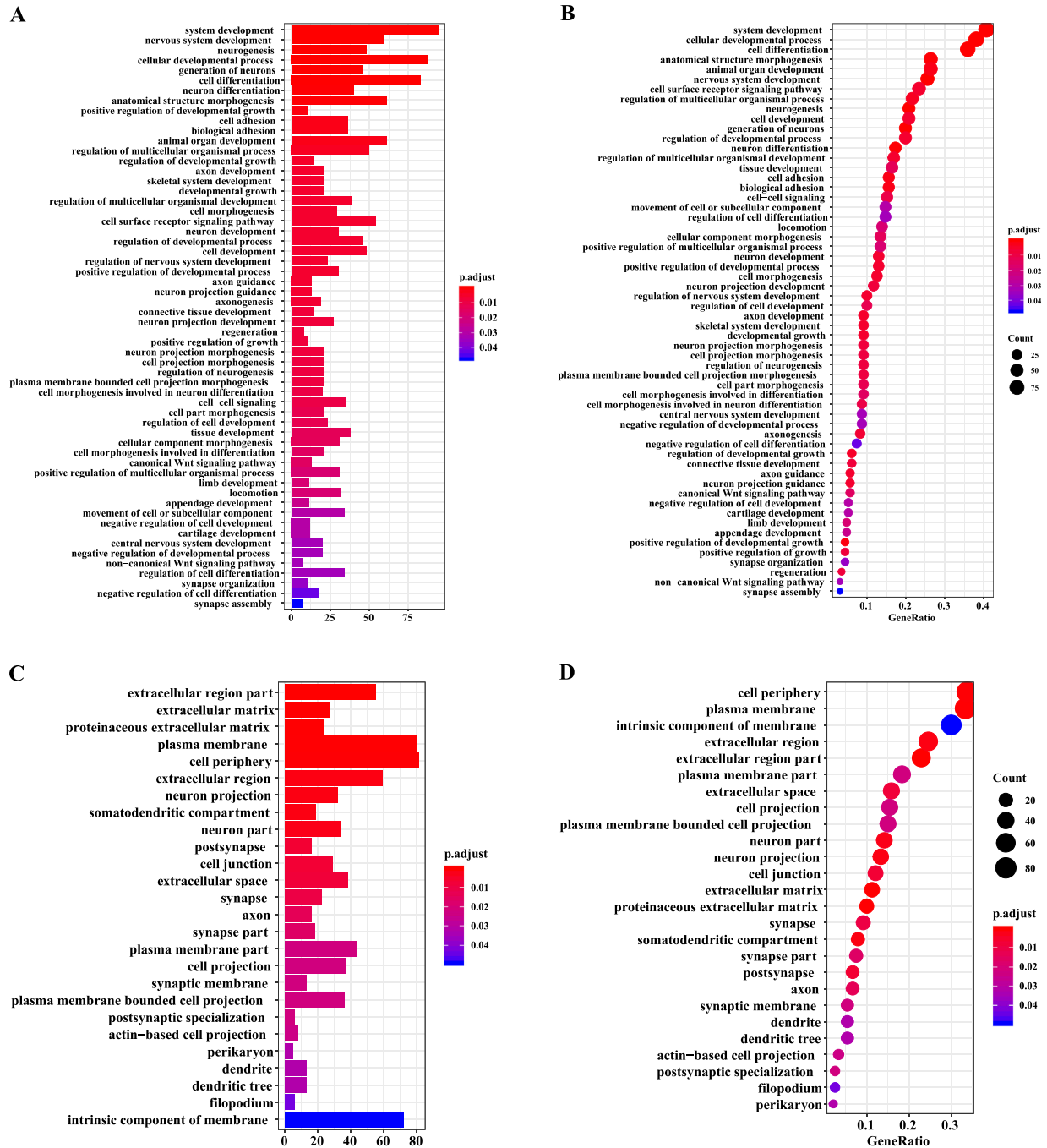


Figure 6. Gene ontology (GO) enrichment of DEGs. (A) Bar chart of GO enrichment in biological process. (B) Scatterplot of GO enrichment in biological process. (C) Bar chart of GO enrichment in cellular component. (D) Scatterplot of GO enrichment in cellular component.

PPI Network Analysis of DEGs In order to better understand the biological relationship between genes, PPI analysis was performed on 13 KEGG pathway DEGs using STRING database (Figure 8). Found *BMPR1A*, *BMP2*, *BMP5*, *BMP6*, *BMP7*, *BMP8A*, *GDF5*, *GDF6*, *SMAD6*, *SMAD9*, *CREBBP*, *LTBP1*, *NFATC1*, *COMP*, *COL6A1*, *COL4A5*, *SPP1*, *ITGA3*, *ITGA7*, *ITGA8*, *ITGA9*, *ITGAV*, *ITGB3*, *ITGB5*, *ITGB8*, *FGFR1*, *FGFR3*, *WNT6*, *WNT7B*, *WNT8A*, *WNT9A*, *WNT9B*, *WNT16*, *FZD1*, *FZD6*, *FZD9*, *FZD10*, *IGF1*, *CAMK4* and other genes are more associated with other genes.

qRT-PCR Validation of DEGs To verify the results of RNA-Seq analysis, 39 selected DEGs were validated by

qRT-PCR (Figure 9). *BMPR1A*, *BMP2*, *BMP5*, *BMP6*, *BMP7*, *BMP8A*, *GDF5*, *SMAD6*, *SMAD9*, *ITGA3*, *ITGA7*, *ITGA8*, *ITGA9*, *ITGAV*, *ITGB3*, *ITGB5*, *ITGB8*, *FGFR1*, *FGFR3*, *WNT6*, *FZD1*, *FZD6*, *FZD10*, *COL6A1*, *COL4A5*, *CREBBP*, *LTBP1*, *NFATC1*, *COMP* and *SPP1* were upregulated in the CMF3 group, and *GDF6*, *IGF1*, *WNT9B*, *WNT16* were downregulated in the CMF3 group. The trends of relative mRNA levels of the above 35 DEGs were consistent with the RNA-Seq data. However, the trends of relative mRNA levels of 5 DEGs, including *FZD9*, *CAMK4*, *WNT7B*, *WNT8A* and *WNT9A*, were inconsistent with the transcriptome analysis results.

Table 4. The Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment of differentially expressed genes (DEGs).

ID	KEGG pathway	P value	P adjust	q value	Count
gga04916	Melanogenesis	0.000002	0.000170	0.000144	25
gga04310	Wnt signaling pathway	0.000003	0.000170	0.000144	33
gga04512	ECM-receptor interaction	0.000017	0.000750	0.000636	22
gga04150	mTOR signaling pathway	0.000155	0.004317	0.003664	29
gga04350	TGF-beta signaling pathway	0.000167	0.004317	0.003664	21
gga04261	Adrenergic signaling in cardiomyocytes	0.000329	0.007067	0.005998	26
gga04933	AGE-RAGE signaling pathway in diabetic complications	0.000475	0.008746	0.007422	20
gga04010	MAPK signaling pathway	0.000596	0.009611	0.008156	43
gga04510	Focal adhesion	0.000911	0.013062	0.011085	33
gga04520	Adherens junction	0.001466	0.018912	0.016049	16
gga04080	Neuroactive ligand-receptor interaction	0.002056	0.024108	0.020459	50
gga04371	Apelin signaling pathway	0.003302	0.035501	0.030127	22
gga04514	Cell adhesion molecules (CAMs)	0.004524	0.044890	0.038095	22

DISCUSSION

Recent studies have shown that the Chinese herbs used in this study have particular effects on bone growth and development and bone protection (Table 5). Therefore, we combined these herbs in different proportions and then added them to the diet of broilers to explore the effects of different CMFs on the bones of broilers. We first assessed the effects of CMFs on the growth performance of broilers. Previous studies have shown that many Chinese herbs could improve the growth performance of broilers, such as *Astragalus*, *Lycium barbarum*, *Chinese yam*, *Salvia*, and others, and these Chinese herbs are rich in polysaccharides, which can significantly increase live weight and daily gain of broilers and reduce the ratio of feed and meat when added into the diet (Long et al., 2020; Han et al., 2021; Qiao et al., 2022; Chang et al., 2023; Zhang et al., 2023). In this study, the results of body weight, ADG, ADFI and FCR showed that CMF supplementation had no adverse effects on broilers.

Bone is an important organ of vertebrates, which plays the role of movement, support, protection and mineral storage. It is extremely important for the growth and development of animals. Bone microstructure and bone biomechanical properties are important indicators to evaluate bone health. Micro-CT is a high-magnification imaging modality that offers the advantages of reproducibility and accuracy, allowing 3D microstructural assessment of trabecular and cortical bone in human and animal bone specimens, and quantifying many skeletal structural parameters with high accuracy

(Feldkamp et al., 1989; Foessl et al., 2021). As a result, Micro-CT is considered the “gold standard” for assessing bone health (Feldkamp et al., 1989). Bone structure parameters such as BMD, BV/TV, Tb.N, Tb.Th, Tb.Sp, etc., can effectively evaluate bone mass and bone quality of bone tissue (Qiu et al., 2019; Chen and Kim, 2020). BMD, or bone mineral density, is a major indicator of bone mass and a common indicator of osteoporosis, which is usually associated with the incidence of fracture (Kobayashi et al., 2020). BV/TV refers to the ratio of bone volume to total volume, can directly reflect the bone mass as well as BMD. BS/TV refers to bone surface area density, which indirectly reflects bone mass. Tb.N refers to the average number of trabeculae per unit length, Tb.Th refers to the average thickness of trabeculae and Tb.Sp refers to the average distance between trabeculae, all 3 are the main indicators for evaluating the spatial morphological structure of trabeculae. All these parameters can reflect bone quality. When bone injury occurs, BV/TV, Tb.N and Tb.Th are significantly decreased, while Tb.Sp is significantly increased (Gielkens et al., 2008).

In recent years, MicroCT has been widely used in the study of small animal bone health, but its use in the domestic chicken bone has been less studied. In this study, the 3D structure diagram of femoral cancellous bone showed that compared with the CON group, the trabecular bone of the distal femur in the CMF groups retained a plate-like structure, the trabecular space was narrowed, and the intertrabecular connections were increased. The results of bone structure parameters showed that BMD, BV/TV, BS/TV, Tb.N, and Tb.Th of femur of broilers in CMF3 groups were increased ($P < 0.05$), while Tb.Sp was decreased ($P < 0.05$) compared with CON group. Previous studies have shown that Chinese herbs such as *Angelica sinensis*, *Chinese yam*, *Salvia miltiorrhiza*, and *Prepared Radix Rehmanniae* could improve BMD and bone microstructure (Table 5). Certain herbal formulations have similar effects. Tang et al. (2020b) found that EF&LLF, a combination of *Epimedium Folium* and *Ligustri Lucidi Fructus*, had bone protective properties and could improve bone mass and quality by preventing bone loss and preserving the microstructure of trabecular bone in osteoporotic rats. He et al. (2022) showed that Bizhongxiao Decoction (BZXD) could effectively alleviate bone damage caused by arthritis by increasing BV/TV, Tb.N and decreasing BS/BV and bone trabecular pattern factor. Liu et al. (2022b) found that Bu-Gu-Sheng-Sui Decoction (BGSSD) has the function of promoting bone formation and improving bone microstructure, which could prevent and treat osteoporosis. It could be seen that the Chinese herbal compounds present the effect of improving bone microstructure. The results of this study show that CMF3 could improve the bone mass of broilers by increasing BMD, BV/TV, BS/TV, Tb.N, Tb.Th and reducing Tb.Sp to increase the trabecular connectivity and protect the microstructure of bone trabeculae.

Biomechanical property of bone is a comprehensive study of bone structure and bone mass, which is a reliable

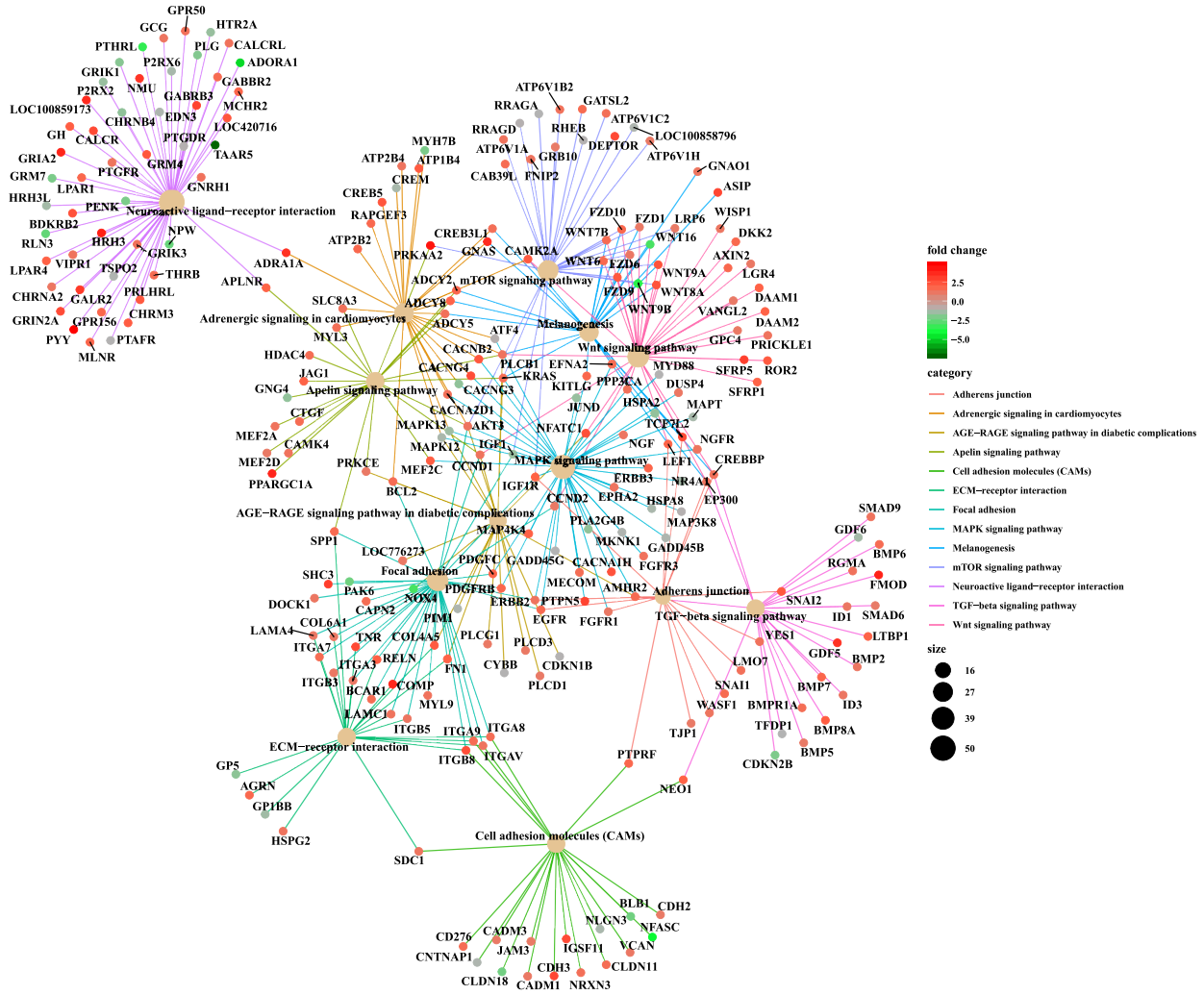


Figure 7. KEGG pathway enrichment of DEGs.

method to evaluate bone mass and can reflect bone strength and determine fracture risk (Jepsen and Schlecht, 2014; Pathria et al., 2016). The 3-point bending test, which is simple and easy to implement, is the first choice for small animal bone biomechanical test, mainly to detect the biomechanical characteristics of bone tissue against external forces (Jepsen et al., 2015; Proding et al., 2018). In this study, the 3-point bending test was performed, and the results showed that compared with the CON group, the elastic modulus, maximum stress, fracture stress ($P < 0.05$) and yield strength ($P > 0.05$) of the tibia of broilers in CMF3 group were increased. The results indicated that CMF3 could improve the biomechanics of tibia bone and reduce the possibility of tibia fracture due to external force. Sun et al. (2020) found that Ching'e Decoction, consisting of *Eucommia ulmoides*, *Psoralea corylifolia* L, *Semen Juglandis*, and *Allium sativum* L, slowed bone loss, improved bone quality, and increased maximum stress and maximum load on the tibia. Liu et al. (2022d) showed that Chinese herbal formula could improve the bone characteristics of broilers, including the improvement of bone microstructure and the enhancement of bone strength. The results of this study were similar to those studies. Studies have also shown that bone biomechanical properties were

related to bone microstructure, and low bone density and deterioration of bone structure may lead to reduced bone biomechanical strength (Dalle Carbonare and Giannini, 2004; Jiang et al., 2007), these conclusions were consistent with our study on bone biomechanics and bone microstructure. These results suggest that CMF3 supplementation may strengthen bone by reducing BMD loss and improving bone microstructure. Although there is evidence that CMF3 has a beneficial effect on bone protection in broilers, further research is needed to obtain more detailed information on the effects of CMF3 on bone protection in broilers.

In summary, the results showed that the CMF3 used in our study could significantly improve the bone microstructure and bone strength of broilers.

Over the years, Chinese herbal medicine has been shown to be effective in the treatment of bone diseases in humans and other mammals. However, there are few studies on the chicken bone and the mechanism is unclear. In the study, transcriptome analysis was performed on broiler bone tissue in CMF3 and CON groups. A total of 2,177 DEGs were identified, including 1,450 upregulated genes and 727 downregulated genes.

Through GO functional annotation and enrichment analysis, the DEGs obtained by analysis could be

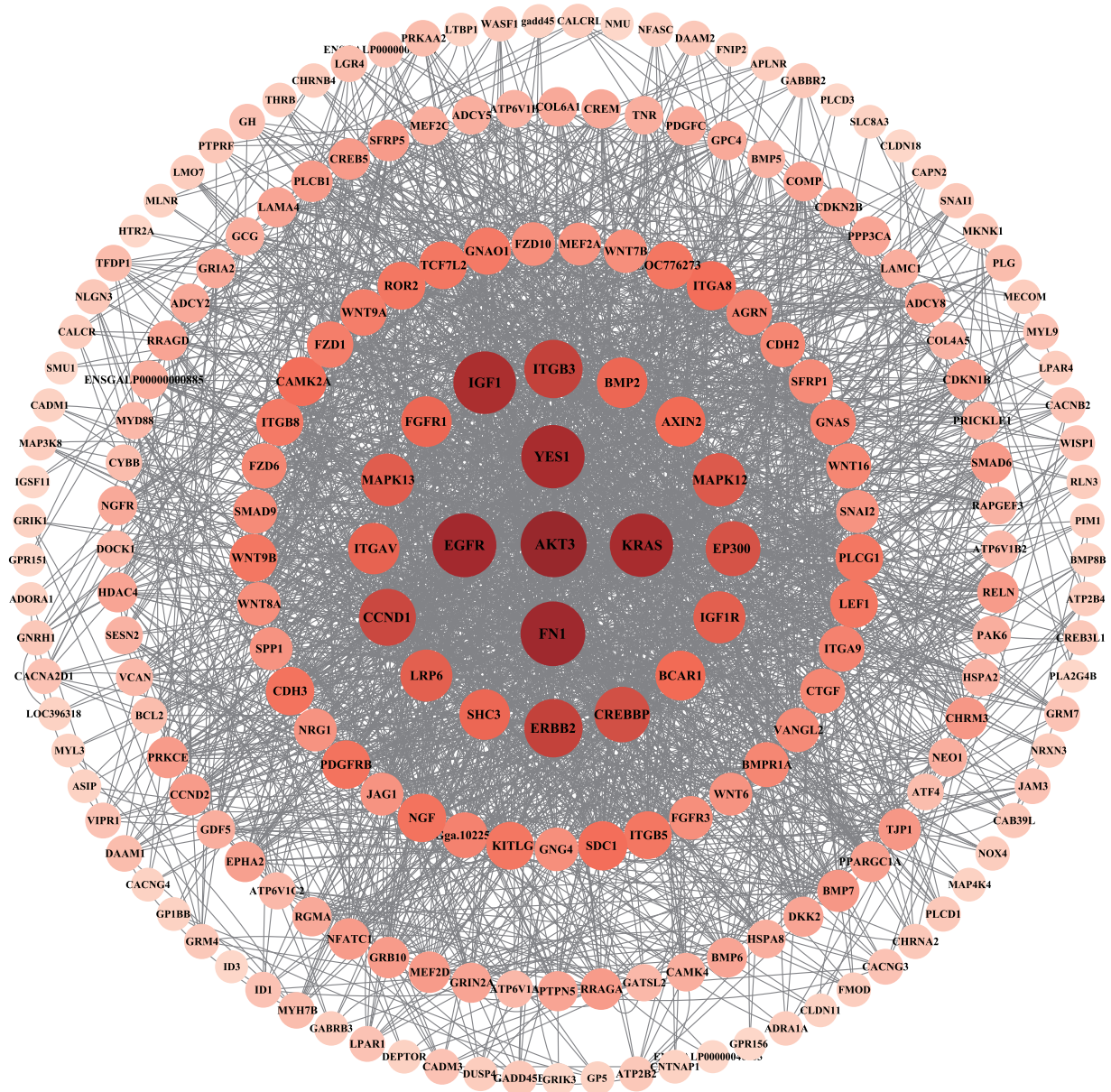


Figure 8. Protein-protein interaction (PPI) network analysis of DEGs.

enriched in the process of cell differentiation and skeletal phylogeny, and are related to bone composition. By matching DEGs to the KEGG database, we identified 13 KEGG pathways including the melanogenesis, Wnt, ECM-receptor interaction, TGF-beta, and mTOR signaling pathway and others. Wnt signaling pathways could be classified as noncanonical and canonical pathways that play a critical role in controlling bone development and homeostasis in vivo (Chen et al., 2019a). The canonical Wnt signaling pathway is a key signaling pathway for maintaining homeostasis in bone tissue, affecting the proliferation and differentiation of mesenchymal stem cells (MSC) and osteoblast progenitors, as well as bone resorption of osteoclasts (Baron and Kneisel, 2013). The noncanonical Wnt signaling pathway is also involved in the regulation of osteoblast differentiation and promotes bone regeneration (Day et al., 2005; Chang et al., 2007). The TGF-beta signaling pathway is one of the key pathways for maintaining bone

homeostasis. It plays a dual role in the process of osteogenesis, not only inducing early differentiation and matrix formation of osteoblasts, but also preventing terminal differentiation and matrix mineralization, which can well maintain bone homeostasis (MacFarlane et al., 2017; Grafe et al., 2018). In addition, the TGF-beta signaling pathway mediates the differentiation of MSC to osteoblasts, chondrocytes, tendon cells, myogenic cells, and adipocytes (Grafe et al., 2018). The MAPK signaling pathway has been shown to play a positive regulatory role in chondrogenesis, osteogenic differentiation, and osteoclast differentiation, and is critical for the regulation of bone development and homeostasis (Yoon et al., 2000; Xiao et al., 2020; Yang et al., 2023). The ECM-receptor interaction pathway plays an important role in bone homeostasis through 3 ECM proteins, namely laminin, fibronectin, and collagen. Studies have shown that ECM-receptor interaction could regulate bone metabolism by influencing the function of

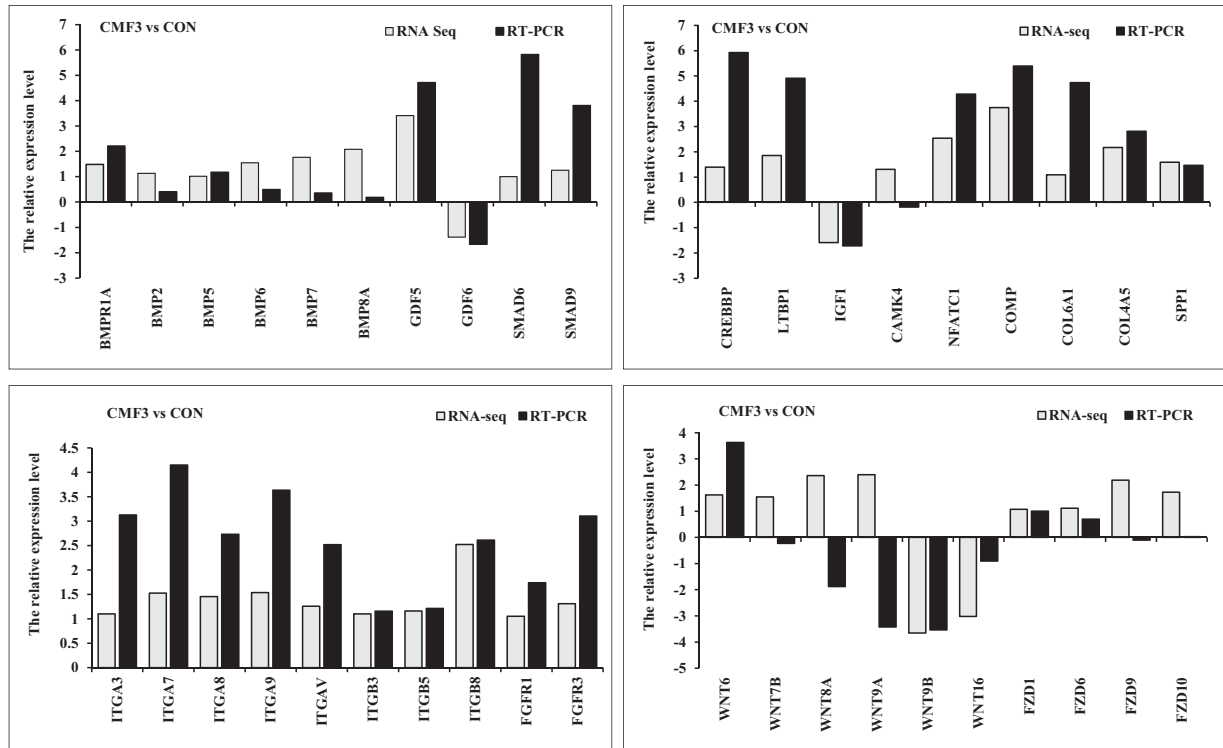


Figure 9. The mRNA expression levels of selected DEGs were evaluated and compared by RNA-seq and RT-PCR ($n = 5$). *BMP1A*, bone morphogenetic protein receptor type-1A; *BMP2*, bone morphogenetic protein 2; *BMP5*, bone morphogenetic protein 5; *BMP6*, bone morphogenetic protein 6; *BMP7*, bone morphogenetic protein 7; *BMP8A*, bone morphogenetic protein 8A; *GDF5*, growth/differentiation factor 5; *GDF6*, growth/differentiation factor 6; *SMAD6*, mothers against decapentaplegic homolog 6; *SMAD9*, mothers against decapentaplegic homolog 9; *CREBBP*, CREB-binding protein; *LTBP1*, latent-transforming growth factor beta-binding protein 1; *IGF1*, insulin-like growth factor 1; *CAMK4*, calcium/calmodulin-dependent protein kinase IV-like; *NFATC1*, nuclear factor of activated T-cells, cytoplasmic 1; *COMP*, cartilage oligomeric matrix protein; *COL6A1*, collagen type VI alpha 1 chain; *COL4A5*, collagen type IV alpha 5 chain; *SPP1*, secreted phosphoprotein-1, osteopontin; *ITGA3*, integrin alpha-3; *ITGA7*, integrin alpha-7; *ITGA8*, integrin alpha-8; *ITGA9*, integrin alpha-9; *ITGAV*, integrin alpha-V; *ITGB3*, integrin beta-3; *ITGB5*, integrin beta-5; *ITGB8*, integrin beta-8; *FGFR1*, fibroblast growth factor receptor 1; *FGFR3*, fibroblast growth factor receptor 3; *WNT6*, Wnt family member 6; *WNT7B*, Wnt family member 7B; *WNT8A*, Wnt family member 8A; *WNT9A*, Wnt family member 9A; *WNT9B*, Wnt family member 9B; *WNT16*, Wnt family member 16; *FZD1*, frizzled class receptor 1; *FZD6*, frizzled class receptor 6; *FZD9*, frizzled class receptor 9; *FZD10*, frizzled class receptor 10.

osteoclasts, and thus affect the bone microenvironment (Shi et al., 2022). The mTOR signaling pathway plays a role in bone size and quality control, regulating bone length and width primarily by coordinating MSC proliferation and differentiation, and increasing bone mineral content by inhibiting osteoclast activity and differentiation (Wu et al., 2017).

Cell adhesion molecules are strongly related to the regulation of regeneration, differentiation and development of hematopoietic stem and progenitor cells, and are indispensable in the bone marrow microenvironment (Klamer and Voermans, 2014). The Adherens junction is called an adhesion belt if it involves a cell-cell connection, or a focal adhesion if it is a cell-extracellular matrix connection. The adherens junction plays a key role in bone development and homeostasis (Marie et al., 2014). It has been reported that Focal adhesion pathway may promote proliferation, development, antiapoptosis, cell survival and cell motility of broiler chondrocytes through Wnt or MAPK signaling pathways (Tian et al., 2013). It may also play a role in the progression of osteoarthritis together with the ECM-receptor interaction pathway (He et al., 2016; Wang et al., 2016). Apelin signaling pathway is critical for a variety of biological functions, including

bone healing, neuroinflammation, liver fibrosis and cirrhosis (Hang et al., 2019; Owen et al., 2021; Liu et al., 2022c). The Neuroactive ligand-receptor interaction pathway has been shown to be potentially involved in osteoclast formation, associated with osteoporosis (Xiao and Zhao, 2019). Adrenergic signaling in cardiomyocytes plays a role in bone remodeling, mainly involving pro-osteoclast gene expression and osteoclast formation (Jiao et al., 2015). AGE-RAGE signaling pathway in diabetic complications has a more complex function in bone metabolism, which seems to have a certain osteoprotective effect as well as negative effect on osteoblast differentiation and function (Zhou and Xiong, 2011; Asadipooya and Uy, 2019). Melanogenesis is a biosynthetic pathway that produces melanin in melanocytes and is essential for the synthesis of melanin. However, no studies have yet identified melanogenesis signaling involved in the regulation of skeletal development. However, in our study, melanogenesis was the top pathway, and whether it is involved in the regulation of skeletal growth and development by CMF3 or plays any role in the skeleton needs to be further investigated.

Normal bones are in a metabolic balance between osteoblasts forming new bone and osteoclasts absorbing

Table 5. Chinese herbs that affect bone growth, development, and health.

Name	Active components	Effects on bone characteristics	References
<i>Angelica sinensis</i>	Z-Ligustilide	Increase BMD of femur.	Lim and Kim (2014)
<i>Lycium barbarum</i>	Polysaccharide	Reduce the degeneration of cartilage matrix, maintain the integrity of articular cartilage.	Ni et al. (2021)
<i>Chinese yam</i>	Aqueous extract	Increased tibial BMD and BV/TV, inhibited osteopenia.	Zhang et al. (2014)
<i>Salvia miltiorrhiza</i>	Tanshinone-IIA	Decrease the incidence of TD and improve the development of tibial cartilage in TD chickens; promote MSC differentiation, increase BMD, bone strength and fracture healing, prevent bone loss.	Mehmood et al. (2017), Zhang et al. (2020a)
<i>Schisandra chinensis</i>	Schisandrin B	Inhibited chondrocytes inflammation in vitro and ameliorated cartilage degeneration in vivo via suppression of NF- κ B and MAPK signal pathways.	Ran et al. (2018)
<i>Rhizoma corydalis</i>	/	Reduce fracture risk.	Chen et al. (2019b)
<i>Fructus cnidii</i>	Osthole	Restore the growth plate size, significantly upregulating the expression of <i>BMP2</i> and <i>RUNX2</i> , alleviate Tibial dyschondroplasia (TD) in chickens.	Waqas et al. (2019)
<i>Oyster</i>	Hydrolysate	Activate the expression of osteogenic genes (<i>RUNX2</i> , <i>ALP</i> , <i>Col1α1</i> , <i>OCN</i> , <i>OSX</i> , and <i>BMP2/4</i>), enhance osteogenesis via crosstalk with the canonical Wnt/ β -catenin pathway.	Molagoda et al. (2019)
<i>Pueraria Lobata</i>	Puerarin	Reduce cell apoptosis, enhance angiogenesis, and restore the width of growth plate, alleviate lameness of TD chickens.	Waqas et al. (2020)
<i>Prepared Radix Rehmanniae</i>	Combined with Radix Astragali	Improve bone microstructure and bone metabolism imbalance, increase BMD.	Kang et al. (2023)
<i>Epimedium</i>	Icariin	Increase the vascular area of growth plate, upregulate the expression of <i>BMP2</i> , and restore TD affected chickens.	Iqbal et al. (2018)
<i>Rhizoma Cyperi</i>	α -Cyperone	Downregulating NF- κ B and MAPKs signaling, reducing chondrocyte inflammation and ECM degradation.	Zhang et al. (2021)
<i>Fructus Psoraleae</i>	Psoralen	The expression of <i>BMP2</i> and <i>BMP4</i> genes is upregulated, and BMP signaling pathway is activated to promote the differentiation of osteoblasts.	Tang et al. (2011)
<i>Rhizoma Drynariae</i>	Total flavonoids	Promote bone growth, reduce claudication, gain weight in TD chickens, and restore growth plate width.	Xu et al. (2022b)
<i>Achyranthes bidentata</i>	β -Ecdysterone	Promote proliferation and induce osteogenic differentiation of bone progenitors through the BMP2/Smad/Runx2/Osterix signaling pathway.	Yan et al. (2022)
<i>Astragalus</i>	Astragaloside IV	Promote the viability and osteogenic ability of BMSCs.	Wang et al. (2021)

old bone. Therefore, maintaining the balance of bone metabolism and protecting the bone microenvironment are the key to healthy bone growth and development. Studies showed that Puerarin, Icariin, total flavonoids of *Rhizoma drynariae*, Psoralen, *Astragalus* polysaccharide, Naringin, and other herbal substances could regulate some of the above signaling pathways to play the role of bone homeostasis regulation (Xiao et al., 2020; Zhang et al., 2020b; Huang et al., 2021; Li et al., 2021; Yang et al., 2021). Our results had some relevance to these findings. In summary, the KEGG pathway analysis in this study indicates that CMF3 may play a role in bone regulation through mediating the conduction of multiple signaling pathways, mainly involving bone formation, chondrogenesis, bone resorption, bone microenvironment regulation, and other processes.

Based on KEGG pathways and PPI analysis, we identified 39 DEGs involved in bone growth and development

(Figures 8 and 9). When the PPI network was used to confirm protein interactions, it was revealed that these 39 genes were associated with other genes and may play an important role in bone function (Figure 8). Among these genes, members of bone morphogenetic protein family (*BMPR1A*, *BMP2*, *BMP5*, *BMP6*, *BMP7*, *BMP8A*), growth differentiation factors (*GDF5*, *GDF6*), and SMAD proteins (*SMAD6*, *SMAD9*) belong to transforming growth factor β (TGF- β) family. BMP is very important for bone metabolism. *BMPR1A* is one of the receptors of BMP, and *BMP2*, 5, 6, and 7 are known as strong osteogenic factors and also play an important role in maintaining cartilage homeostasis and osteoclast differentiation (Suttapreyasri et al., 2006; Huang et al., 2010; Huntley et al., 2019). *BMP8A* is involved in bone formation (Passaro et al., 2021). As ligands to BMP, *GDF5* and *GDF6* are required for bone and joint formation, with mutations in *GDF5* cause developmental defects in

the wrist and ankle, and inactivation of *GDF6* causes defects in joints, ligaments, and chondrogenesis (Settle et al., 2003). SMAD protein mediates TGF- β signaling. Studies have shown that *SMAD9* could inhibit osteoblast activity, while the absence of *SMAD6* would lead to bone defects (Gregson et al., 2020; Estrada et al., 2011).

WNT ligands are highly conserved secreted glycoproteins that have been extensively studied in bone function and disease. *WNT6*, as a regulator of MSC fate, has been shown to inhibit lipogenic differentiation of MSC and promote osteogenic differentiation (Cawthorn et al., 2012). *WNT7B* is an effective bone anabolic protein during embryonic development and after birth, which can significantly increase the number and function of osteoblasts and effectively enhance bone formation (Chen et al., 2014a). *WNT8A* has been shown to be associated with osteoporotic hip fractures (Garcia-Ibarbia et al., 2013). The role of *WNT9A* in bone is mainly related to chondrocyte regulation, which is necessary to maintain joint integrity (Spater et al., 2006; Ling et al., 2017).

WNT9B was related to femur BMD and was considered to be a susceptibility gene for bone strength (Karasik et al., 2016). Other studies have shown that *WNT9B* is not related to bone differentiation (Witte et al., 2009).

WNT16 is derived from osteoblasts, and as a key factor to inhibit the activity of osteoclasts, its absence can lead to decreased bone mass and spontaneous fracture of cortical bone in mice (Moverare-Skrtic et al., 2014). Frizzled class receptor (FZD) is a receptor for the WNT pathway whose role in bone development has been demonstrated. It has been reported that *FZD1* may contribute to the genetic regulation of bone mass and geometry via the WNT pathway (Yerges et al., 2009). *FZD6* plays an important role in regulating the osteogenic ability of osteoporosis mice in vitro and in vivo, and its overexpression promotes the osteogenic ability of osteoporosis mice (Wu et al., 2021). *FZD9* is differentially expressed in the early stage of osteoblast differentiation, and lack of *FZD9* leads to impaired bone formation and low bone mass (Albers et al., 2011). *FZD10* may drive chondrocyte hypertrophy and maturation (Paradise et al., 2018). *COL6A1* and *COL4A5*, members of the solute vector family, are upregulated in the CMF3 group. *COL6A1* has an important role in regulating the physiology of synovial joints, with *COL6A1*^(-/-) mice showing significantly reduced mechanical properties and accelerating the development of arthritic joint degeneration (Alexopoulos et al., 2009). Increased *COL4A5* levels may improve bone formation function (Twine et al., 2018).

Integrin plays an important role in regulating cell proliferation and apoptosis. *ITGA3*, *ITGAV*, and *ITGB5* were all associated with the development of osteosarcoma (Yang et al., 2014; Zheng et al., 2016; Li et al., 2022b). *ITGA7* is associated with bone formation and its overexpression improves osteogenesis through the PI3K/Akt pathway, while its suppression promotes osteoclast activity and accelerates bone loss (Huang et al., 2020; Liu et al., 2022a). *ITGA8* is associated with craniofacial bone development and its mutation leads to

skeletal defects (Talbot et al., 2016). *ITGA9* has been reported to mediate the interaction between bone microenvironment and prostate cancer (San Martin et al., 2017). *ITGB3* overexpression can promote osteogenic differentiation by promoting the formation of mineralized nodules, increasing the activity of ALP and the expression of *RUNX2*, *SPP1*, and ALP in MSCs (Yu et al., 2022). *ITGB8* is found in chondrocyte sheets and chondrocyte sheets-bone tissue and may play a key role in chondrocyte sheet attachment and cartilage ECM synthesis (Wongin et al., 2021). *FGFR1* and *FGFR3*, members of the fibroblast growth factor receptor (FGFR) family, negatively regulate long bone growth and endochondral ossification, respectively, and activate mutations that both lead to bone diseases (White et al., 2005; Zhou et al., 2015).

Other DEGs, *CREBBP* is critical for the regulation of the bone marrow microenvironment and haploinsufficiency of *CREBBP* leads to alterations in the bone marrow microenvironment, mainly manifested as obvious reduction of bone trabecula (Zimmer et al., 2011). As an important extracellular matrix protein, *LTBP1* may directly affect chondrocytes, and its absence of function may lead to severe mandibular defects (Xiong et al., 2020). *NFATC1* regulates bone mass by acting on osteoblasts and osteoclasts (Winslow et al., 2006). As an important extracellular matrix protein, *COMP* plays an important role in the endochondral ossification (Burger et al., 2020). *SPP1* supports osteogenic differentiation of MSC and is an important factor in regulating the balance between fat and bone mass (Chen et al., 2014b). *IGF1* regulates all stages of chondrocyte differentiation in growth plates, and it has also been shown that excess *IGF1* inhibits bone formation (Hunziker et al., 1994; Huang et al., 2010). *CaMK4* is important in the development of arthritis, and inhibition of *CaMK4* can significantly reduce the number of osteoclasts differentiated from CD14 cells in vitro and control the development of arthritis in mice (Koga et al., 2023). It is evident that these 39 genes play important role in bone and cartilage formation. Our research showed that trends of relative mRNA levels of *BMPR1A*, *BMP2*, *BMP5*, *BMP6*, *BMP7*, *BMP8A*, *GDF5*, *SMAD6*, *SMAD9*, *ITGA3*, *ITGA7*, *ITGA8*, *ITGA9*, *ITGAV*, *ITGB3*, *ITGB5*, *ITGB8*, *FGFR1*, *FGFR3*, *WNT6*, *FZD1*, *FZD6*, *FZD10*, *COL6A1*, *COL4A5*, *CREBBP*, *LTBP1*, *NFATC1*, *COMP*, *SPP1* (upregulated by CMF3), and *GDF6*, *IGF1*, *WNT9B*, *WNT16* (downregulated by CMF3) were consistent with the RNA-Seq. The study suggests that CMF3 may regulate bone formation, cartilage formation, and bone resorption in broilers by regulating the expression of these genes. However, the relative mRNA level trends of 5 DEGs, including *FZD9*, *CAMK4*, *WNT7B*, *WNT8A*, and *WNT9A*, were inconsistent with the results of RNA-Seq. Therefore, the roles of these 5 genes in the skeleton and the mechanism of their regulation by CMF3 need to be further investigated to provide more detailed information.

The results of this study indicate that feeding CMF3 could increase the expression of bone and cartilage genes

in broilers. It is speculated that CMF3 may maintain the dynamic balance process of bone formation and bone resorption by regulating the expression of bone-related genes, thus improving bone microstructure and enhancing bone mass and strength.

CONCLUSIONS

In conclusion, CMF3 could improve the bone strength and microstructure of broilers by maintaining the dynamic balance process of bone formation and bone resorption by regulating the expression of bone-related genes, thus improving bone microstructure and enhancing bone mass and strength. It can be seen that, CMF3 could improve bone quality in broilers, and Wnt signaling pathway, TGF- β signaling pathway, MAPK signaling pathway, mTOR signaling pathway, ECM-receptor interaction, and other signaling pathways were the important pathways for bone formation, and regulated by the genes involved in these pathways. Although our research has achieved positive results, there are still many questions to be explored and resolved. In future studies, we may explore the regulatory mechanism of CMF on bone growth and development from in vitro cell experiments, to identify genes and signal transduction pathways with unknown functions. Finally, we believe that more scientific, standardized, and reasonable research and practice of drugs still need to continue the efforts of the world.

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DISCLOSURES

The authors declare that they have no competing interests.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.psj.2023.103044](https://doi.org/10.1016/j.psj.2023.103044).

REFERENCES

- Albers, J., J. Schulze, F. T. Beil, M. Gebauer, A. Baranowsky, J. Keller, R. P. Marshall, K. Wintges, F. W. Friedrich, M. Priemel, A. F. Schilling, J. M. Rueger, K. Cornils, B. Fehse, T. Streichert, G. Sauter, F. Jakob, K. L. Insogna, B. Pober, K. P. Knobloch, U. Francke, M. Amling, and T. Schinke. 2011. Control of bone formation by the serpentine receptor Frizzled-9. *J. Cell. Biol.* 192:1057–1072.
- Alexopoulos, L. G., I. Youn, P. Bonaldo, and F. Guilak. 2009. Developmental and osteoarthritic changes in Col6a1-knockout mice: biomechanics of type VI collagen in the cartilage pericellular matrix. *Arthritis Rheum.* 60:771–779.
- Asadipooaya, K., and E. M. Uy. 2019. Advanced glycation end products (AGEs), receptor for AGEs, diabetes, and bone: review of the literature. *J. Endocr. Soc.* 3:1799–1818.
- Baron, R., and M. Kneissel. 2013. WNT signaling in bone homeostasis and disease: from human mutations to treatments. *Nat. Med.* 19:179–192.
- Burger, A., J. Roosenboom, M. Hossain, S. M. Weinberg, J. T. Hecht, and K. L. Posey. 2020. Mutant COMP shapes growth and development of skull and facial structures in mice and humans. *Mol. Genet. Genomic Med.* 8:e1251.
- Cawthorn, W. P., A. J. Bree, Y. Yao, B. Du, N. Hemati, G. Martinez-Santibanez, and O. A. MacDougald. 2012. Wnt6, Wnt10a and Wnt10b inhibit adipogenesis and stimulate osteoblastogenesis through a beta-catenin-dependent mechanism. *Bone* 50:477–489.
- Chang, J., W. Sonoyama, Z. Wang, Q. Jin, C. Zhang, P. H. Krebsbach, W. Giannobile, S. Shi, and C. Y. Wang. 2007. Noncanonical Wnt-4 signaling enhances bone regeneration of mesenchymal stem cells in craniofacial defects through activation of p38 MAPK. *J. Biol. Chem.* 282:30938–30948.
- Chang, Y., J. Zhang, Y. Jin, J. Deng, M. Shi, and Z. Miao. 2023. Effects of dietary supplementation of chinese yam polysaccharide on carcass composition, meat quality, and antioxidant capacity in broilers. *Animals (Basel)* 13:503.
- Chen, C., and W. K. Kim. 2020. The application of micro-CT in egg-laying hen bone analysis: introducing an automated bone separation algorithm. *Poult. Sci.* 99:5175–5183.
- Chen, W. J., H. Livneh, M. H. Hsieh, C. C. Yeh, M. H. Yeh, M. C. Lu, J. T. Chien, and T. Y. Tsai. 2019b. Association of use of Chinese herbal medicines and the risk of fracture in patients with osteoporosis: a population-based cohort study. *Osteoporos. Int.* 30:807–815.
- Chen, Q., P. Shou, L. Zhang, C. Xu, C. Zheng, Y. Han, W. Li, Y. Huang, X. Zhang, C. Shao, A. I. Roberts, A. B. Rabson, G. Ren, Y. Zhang, Y. Wang, D. T. Denhardt, and Y. Shi. 2014b. An osteopontin-integrin interaction plays a critical role in directing adipogenesis and osteogenesis by mesenchymal stem cells. *Stem Cells* 32:327–337.
- Chen, J., X. Tu, E. Esen, K. S. Joeng, C. Lin, J. M. Arbeit, M. A. Ruegg, M. N. Hall, L. Ma, and F. Long. 2014a. WNT7B promotes bone formation in part through mTORC1. *PLoS Genet* 10: e1004145.
- Chen, D., R. Xie, B. Shu, A. L. Landay, C. Wei, J. Reiser, A. Spagnoli, A. Torquati, C. B. Forsyth, A. Keshavarzian, and D. R. Sumner. 2019a. Wnt signaling in bone, kidney, intestine, and adipose tissue and interorgan interaction in aging. *Ann. N. Y. Acad. Sci.* 1442:48–60.
- Dalle Carbonare, L., and S. Giannini. 2004. Bone microarchitecture as an important determinant of bone strength. *J. Endocrinol. Invest.* 27:99–105.
- Day, T. F., X. Guo, L. Garrett-Beal, and Y. Yang. 2005. Wnt/ β -catenin signaling in mesenchymal progenitors controls osteoblast and chondrocyte differentiation during vertebrate skeletogenesis. *Dev. Cell* 8:739–750.
- Dinev, I. 2012. Clinical and morphological investigations on the incidence of forms of rickets and their association with other pathological states in broiler chickens. *Res. Vet. Sci.* 92:273–277.
- Dou, T., S. Yan, L. Liu, K. Wang, Z. Jian, Z. Xu, J. Zhao, Q. Wang, S. Sun, M. Z. Talpur, X. Duan, D. Gu, Y. He, Y. Du, A. M. Abdulwahid, Q. Li, H. Rong, W. Cao, Z. Su, G. Zhao, R. Liu, S. Zhao, Y. Huang, M. F. W. Te Pas, C. Ge, and J. Jia. 2022. Integrative analysis of transcriptomics and metabolomics to reveal the melanogenesis pathway of muscle and related meat characters in Wuliangshan black-boned chickens. *BMC Genom.* 23:173.
- Estrada, K. D., K. N. Retting, A. M. Chin, and K. M. Lyons. 2011. Smad6 is essential to limit BMP signaling during cartilage development. *J. Bone Miner. Res.* 26:2498–2510.
- Feldkamp, L. A., S. A. Goldstein, A. M. Parfitt, G. Jesion, and M. Kleerekoper. 1989. The direct examination of three-dimensional bone architecture in vitro by computed tomography. *J. Bone Miner. Res.* 4:3–11.

- Foessel, I., J. H. D. Bassett, A. Bjornerem, B. Busse, A. Calado, P. Chavassieux, M. Christou, E. Douni, I. A. K. Fiedler, J. E. Fonseca, E. Hassler, W. Hogler, E. Kague, D. Karasik, P. Khashayar, B. L. Langdahl, V. D. Leitch, P. Lopes, G. Markozannes, F. E. A. McGuigan, C. Medina-Gomez, E. Ntzani, L. Oei, C. Ohlsson, P. Szulc, J. H. Tobias, K. Trajanoska, S. Tuzun, A. Valjevac, B. van Rietbergen, G. R. Williams, T. Zekic, F. Rivadeneira, and B. Obermayer-Pietsch. 2021. Bone phenotyping approaches in human, mice and zebrafish – expert overview of the EU cost action GEMSTONE ("GENomics of MusculoSkeletal traits Translational NETwork"). *Front. Endocrinol. (Lausanne)* 12:720728.
- Garcia-Ibarbia, C., J. Delgado-Calle, I. Casafont, J. Velasco, J. Arozamena, M. I. Perez-Nunez, M. A. Alonso, M. T. Berciano, F. Ortiz, J. L. Perez-Castrillon, A. F. Fernandez, M. F. Fraga, M. T. Zarrabeitia, and J. A. Riancho. 2013. Contribution of genetic and epigenetic mechanisms to Wnt pathway activity in prevalent skeletal disorders. *Gene* 532:165–172.
- Gielkens, P. F., J. Schortinghuis, J. R. de Jong, M. C. Huysmans, M. B. Leeuwen, G. M. Raghoobar, R. R. Bos, and B. Stegenga. 2008. A comparison of micro-CT, microradiography and histomorphometry in bone research. *Arch. Oral Biol.* 53:558–566.
- Grafe, I., S. Alexander, J. R. Peterson, T. N. Snider, B. Levi, B. Lee, and Y. Mishina. 2018. TGF-beta family signaling in mesenchymal differentiation. *Cold Spring Harb. Perspect. Biol.* 10:a022202.
- Gregson, C. L., D. J. M. Bergen, P. Leo, R. B. Sessions, L. Wheeler, A. Hartley, S. Youtlen, P. I. Croucher, A. M. McInerney-Leo, W. Fraser, J. C. Tang, L. Anderson, M. Marshall, L. Sergot, L. Paternoster, G. Davey Smith, A. Consortium, M. A. Brown, C. Hammond, J. P. Kemp, J. H. Tobias, and E. L. Duncan. 2020. A rare mutation in SMAD9 associated with high bone mass identifies the SMAD-dependent BMP signaling pathway as a potential anabolic target for osteoporosis. *J. Bone Miner. Res.* 35:92–105.
- Han, C., X. Wang, D. Zhang, Y. Wei, Y. Cui, W. Shi, and Y. Bao. 2021. Synergistic use of florfenicol and *Salvia miltiorrhiza* polysaccharide can enhance immune responses in broilers. *Ecotoxicol. Environ. Saf.* 210:111825.
- Hang, K., C. Ye, J. Xu, E. Chen, C. Wang, W. Zhang, L. Ni, Z. Kuang, L. Ying, D. Xue, and Z. Pan. 2019. Apelin enhances the osteogenic differentiation of human bone marrow mesenchymal stem cells partly through Wnt/beta-catenin signaling pathway. *Stem Cell Res. Ther.* 10:189.
- Harash, G., K. C. Richardson, Z. Alshamy, H. Hünigen, H. M. Hafez, J. Plendl, and S. Al Masri. 2020. Basic morphometry, microcomputed tomography and mechanical evaluation of the tibiotarsal bone of a dual-purpose and a broiler chicken line. *PLoS One* 15: e0230070.
- He, J., X. Li, Z. Wang, S. Bennett, K. Chen, Z. Xiao, J. Zhan, S. Chen, Y. Hou, J. Chen, S. Wang, J. Xu, and D. Lin. 2019. Therapeutic anabolic and anticatabolic benefits of natural Chinese medicines for the treatment of osteoporosis. *Front. Pharmacol.* 10:1344.
- He, C., Y. Wang, Y. Wen, T. Li, E. Hu, S. Zeng, B. Yang, and X. Xiong. 2022. An intersectional analysis of lncRNAs and mRNAs reveals the potential therapeutic targets of Bi Zhong Xiao Decoction in collagen-induced arthritis rats. *Chin. Med.* 17:110.
- He, P., Z. Zhang, W. Liao, D. Xu, M. Fu, and Y. Kang. 2016. Screening of gene signatures for rheumatoid arthritis and osteoarthritis based on bioinformatics analysis. *Mol. Med. Rep.* 14:1587–1593.
- Huai, Y., W. J. Zhang, W. Wang, K. Dang, S. F. Jiang, D. M. Li, M. Li, Q. Hao, Z. P. Miao, Y. Li, and A. R. Qian. 2021. Systems pharmacology dissection of action mechanisms for herbs in osteoporosis treatment. *Chin. Herb. Med.* 13:313–331.
- Huang, Y., L. Liao, H. Su, X. Chen, T. Jiang, J. Liu, and Q. Hou. 2021. Psoralen accelerates osteogenic differentiation of human bone marrow mesenchymal stem cells by activating the TGF-beta/Smad3 pathway. *Exp. Ther. Med.* 22:940.
- Huang, Z., P. G. Ren, T. Ma, R. L. Smith, and S. B. Goodman. 2010. Modulating osteogenesis of mesenchymal stem cells by modifying growth factor availability. *Cytokine* 51:305–310.
- Huang, T., Z. Yu, Q. Yu, Y. Wang, Z. Jiang, H. Wang, and G. Yang. 2020. Inhibition of osteogenic and adipogenic potential in bone marrow-derived mesenchymal stem cells under osteoporosis. *Biochem. Biophys. Res. Commun.* 525:902–908.
- Huntley, R., E. Jensen, R. Gopalakrishnan, and K. C. Mansky. 2019. Bone morphogenetic proteins: their role in regulating osteoclast differentiation. *Bone Rep.* 10:100207.
- Hunziker, E. B., J. Wagner, and J. Zapf. 1994. Differential effects of insulin-like growth factor I and growth hormone on developmental stages of rat growth plate chondrocytes in vivo. *J. Clin. Invest.* 93:1078–1086.
- Iqbal, M., H. Zhang, K. Mehmood, A. Li, X. Jiang, Y. Wang, J. Zhang, M. K. Iqbal, M. U. Rehman, W. Yao, S. Yang, and J. Li. 2018. Icaritin: a potential compound for the recovery of tibial dyschondroplasia affected chicken via up-regulating BMP-2 expression. *Biol. Proc. Online* 20:15.
- Jepsen, K. J., and S. H. Schlecht. 2014. Biomechanical mechanisms: resolving the apparent conundrum of why individuals with type II diabetes show increased fracture incidence despite having normal BMD. *J. Bone Miner. Res.* 29:784–786.
- Jepsen, K. J., M. J. Silva, D. Vashishth, X. E. Guo, and M. C. van der Meulen. 2015. Establishing biomechanical mechanisms in mouse models: practical guidelines for systematically evaluating phenotypic changes in the diaphyses of long bones. *J. Bone Miner. Res.* 30:951–966.
- Jiang, S. D., L. S. Jiang, and L. Y. Dai. 2007. Changes in bone mass, bone structure, bone biomechanical properties, and bone metabolism after spinal cord injury: a 6-month longitudinal study in growing rats. *Calcif. Tissue Int.* 80:167–175.
- Jiao, K., L. N. Niu, Q. H. Li, G. T. Ren, C. M. Zhao, Y. D. Liu, F. R. Tay, and M. Q. Wang. 2015. beta2-adrenergic signal transduction plays a detrimental role in subchondral bone loss of temporomandibular joint in osteoarthritis. *Sci. Rep.* 5:12593.
- Joiner, K. S., F. J. Hoerr, E. van Santen, and S. J. Ewald. 2005. The avian major histocompatibility complex influences bacterial skeletal disease in broiler breeder chickens. *Vet. Pathol.* 42:275–281.
- Kang, W. Q., P. F. Wei, L. Ou, M. Li, and C. Y. Liu. 2023. The mechanism study of inhibition effect of prepared Radix Rehmanniae combined with Radix Astragali osteoporosis through PI3K-AKT signaling pathway. *Acta Cir. Bras.* 37:e371101.
- Karasik, D., F. Rivadeneira, and M. L. Johnson. 2016. The genetics of bone mass and susceptibility to bone diseases. *Nat. Rev. Rheumatol.* 12:323–334.
- Kieróńczyk, B., M. Rawski, D. Józefiak, and S. Świątkiewicz. 2017. Infectious and non-infectious factors associated with leg disorders in poultry - a review. *Ann. Anim. Sci.* 17:645–669.
- Klamer, S., and C. Voermans. 2014. The role of novel and known extracellular matrix and adhesion molecules in the homeostatic and regenerative bone marrow microenvironment. *Cell Adh. Migr.* 8:563–577.
- Kobayashi, T., M. Kaneko, and M. Narukawa. 2020. Influence of prevalent vertebral fracture on the correlation between change in lumbar spine bone mineral density and risk of new vertebral fracture: a meta-analysis of randomized clinical trials. *Clin. Drug Investig.* 40:15–23.
- Koga, T., M. Umeda, N. Yoshida, A. Satyam, M. Jha, M. Scherlinger, R. Bhargava, M. G. Tsokos, T. Sato, K. Furukawa, Y. Endo, S. Fukui, N. Iwamoto, N. Abiru, M. Okita, M. Ito, A. Kawakami, and G. C. Tsokos. 2023. Inhibition of calcium/calmodulin-dependent protein kinase IV in arthritis: dual effect on Th17 cell activation and osteoclastogenesis. *Rheumatology (Oxford)* 62:861–871.
- Li, Z., L. Deng, Y. Li, and Y. Wang. 2022b. MiR-139 inhibits proliferation, migration and invasion of osteosarcoma cell line MG63 via down-regulating integrin subunit alpha V (ITGAV). *Tissue Cell* 75:101720.
- Li, Y., L. Li, X. Li, B. Luo, Q. Ye, H. Wang, L. Yang, X. Zhu, L. Han, R. Zhang, H. Tian, and P. Wang. 2022a. A mechanistic review of chinese medicine polyphenols on bone formation and resorption. *Front. Pharmacol.* 13:1017538.
- Li, S., H. Zhou, C. Hu, J. Yang, J. Ye, Y. Zhou, Z. Li, L. Chen, and Q. Zhou. 2021. Total flavonoids of *rhizoma drynariae* promotes differentiation of osteoblasts and growth of bone graft in induced membrane partly by activating wnt/beta-catenin signaling pathway. *Front. Pharmacol.* 12:675470.
- Lim, D. W., and Y. T. Kim. 2014. Anti-osteoporotic effects of *Angelica sinensis* (Oliv.) Diels extract on ovariectomized rats and its oral toxicity in rats. *Nutrients* 6:4362–4372.
- Lin, J., J. Zhu, Y. Wang, N. Zhang, H. J. Gober, X. Qiu, D. Li, and L. Wang. 2017. Chinese single herbs and active ingredients for

- postmenopausal osteoporosis: from preclinical evidence to action mechanism. *Biosci. Trends* 11:496–506.
- Ling, I. T., L. Rochard, and E. C. Liao. 2017. Distinct requirements of *wls*, *wnt9a*, *wnt5b* and *gpc4* in regulating chondrocyte maturation and timing of endochondral ossification. *Dev. Biol.* 421:219–232.
- Liu, Y., S. Liang, X. Zi, S. Yan, M. Liu, M. Li, Y. Zhao, T. Dou, C. Ge, K. Wang, and J. Jia. 2022d. Influence of Chinese herbal formula on bone characteristics of Cobb broiler chickens. *Genes (Basel)* 13:1865.
- Liu, K., K. Wang, L. Wang, and Z. Zhou. 2021. Changes of lipid and bone metabolism in broilers with spontaneous femoral head necrosis. *Poult. Sci.* 100:100808.
- Liu, L. L., Y. S. Xiao, W. M. Huang, S. Liu, L. X. Huang, J. H. Zhong, P. Jia, and W. Y. Liu. 2022a. ATF1/miR-214-5p/ITGA7 axis promotes osteoclastogenesis to alter OVX-induced bone absorption. *Mol. Med.* 28:56.
- Liu, N., B. Qi, Y. Zhang, S. Fang, C. Sun, Q. Li, and X. Wei. 2022b. Bu-Gu-Sheng-Sui decoction promotes osteogenesis via activating the ERK/Smad signaling pathways. *Front. Pharmacol.* 13:976121.
- Liu, Q., S. Zhou, X. Wang, C. Gu, Q. Guo, X. Li, C. Zhang, N. Zhang, L. Zhang, and F. Huang. 2022c. Apelin alleviated neuroinflammation and promoted endogenous neural stem cell proliferation and differentiation after spinal cord injury in rats. *J. Neuroinflamm.* 19:160.
- Long, L. N., B. J. Kang, Q. Jiang, and J. S. Chen. 2020. Effects of dietary *Lycium barbarum* polysaccharides on growth performance, digestive enzyme activities, antioxidant status, and immunity of broiler chickens. *Poult. Sci.* 99:744–751.
- MacFarlane, E. G., J. Haupt, H. C. Dietz, and E. M. Shore. 2017. TGF- β family signaling in connective tissue and skeletal diseases. *Cold Spring Harb. Perspect. Biol.* 9:a022269.
- Marie, P. J., E. Hay, D. Modrowski, L. Revollo, G. Mbalaviele, and R. Civitelli. 2014. Cadherin-mediated cell-cell adhesion and signaling in the skeleton. *Calcif. Tissue Int.* 94:46–54.
- Mehmood, K., H. Zhang, M. K. Iqbal, M. U. Rehman, M. Shahzad, K. Li, S. Huang, F. Nabi, L. Zhang, and J. Li. 2017. In vitro effect of apigenin and danshen in tibial dyschondroplasia through inhibition of heat-shock protein 90 and vascular endothelial growth factor expressions in avian growth plate cells. *Avian Dis.* 61:372–377.
- Molagoda, I. M. N., W. A. H. M. Karunarathne, Y. H. Choi, E. K. Park, Y. J. Jeon, B. J. Lee, C. H. Kang, and G. Y. Kim. 2019. Fermented oyster extract promotes osteoblast differentiation by activating the Wnt/ β -Catenin signaling pathway, leading to bone formation. *Biomolecules* 9:711.
- Moverare-Skrtic, S., P. Henning, X. Liu, K. Nagano, H. Saito, A. E. Borjesson, K. Sjogren, S. H. Windahl, H. Farman, B. Kindlund, C. Engdahl, A. Koskela, F. P. Zhang, E. E. Eriksson, F. Zaman, A. Hammarstedt, H. Isaksson, M. Bally, A. Kassem, C. Lindholm, O. Sandberg, P. Aspenberg, L. Savendahl, J. Q. Feng, J. Tuckermann, J. Tuukkanen, M. Poutanen, R. Baron, U. H. Lerner, F. Gori, and C. Ohlsson. 2014. Osteoblast-derived WNT16 represses osteoclastogenesis and prevents cortical bone fragility fractures. *Nat. Med.* 20:1279–1288.
- Ni, J., M. Au, H. Kong, X. Wang, and C. Wen. 2021. *Lycium barbarum* polysaccharides in ageing and its potential use for prevention and treatment of osteoarthritis: a systematic review. *BMC Complement. Med. Ther.* 21:212.
- OECD/FAO. 2020. OECD – FAO Agricultural Outlook 2020–2029. Accessed May 2023. <https://www.oecd-ilibrary.org/sites/1112c23b-en/index.html?itemId=/content/publication/1112c23b-en>.
- Owen, N. E., D. Nyimantu, R. E. Kuc, P. D. Upton, N. W. Morrell, G. J. Alexander, J. J. Maguire, and A. P. Davenport. 2021. Plasma levels of apelin are reduced in patients with liver fibrosis and cirrhosis but are not correlated with circulating levels of bone morphogenetic protein 9 and 10. *Peptides* 136:170440.
- Paradise, C. R., C. Galeano-Garces, D. Galeano-Garces, A. Dudakovic, T. A. Milbrandt, D. B. F. Saris, A. J. Krych, M. Karperien, G. B. Ferguson, D. Evseenko, S. M. Riester, A. J. van Wijnen, and A. N. Larson. 2018. Molecular characterization of physis tissue by RNA sequencing. *Gene* 668:87–96.
- Passaro, D., M. Garcia-Albornoz, G. Diana, P. Chakravarty, L. Ariza-McNaughton, A. Batsivari, C. Borrás-Eroles, A. Abarrategi, A. Wacławiczek, L. Ombrato, I. Malanchi, J. Gribben, and D. Bonnet. 2021. Integrated OMICs unveil the bone-marrow microenvironment in human leukemia. *Cell Rep.* 35:109119.
- Pathria, M. N., C. B. Chung, and D. L. Resnick. 2016. Acute and stress-related injuries of bone and cartilage: pertinent anatomy, basic biomechanics, and imaging perspective. *Radiology* 280:21–38.
- Prodinger, P. M., D. Burklein, P. Foehr, K. Kreutzer, H. Pilge, A. Schmitt, R. V. Eisenhart-Rothe, R. Burgkart, O. Bissinger, and T. Fischer. 2018. Improving results in rat fracture models: enhancing the efficacy of biomechanical testing by a modification of the experimental setup. *BMC Musculoskelet. Disord.* 19:243.
- Qian, D., H. Zhou, P. Fan, T. Yu, A. Patel, M. O'Brien, Z. Wang, S. Lu, G. Tong, Y. Shan, L. Wang, Y. Gao, Y. Xiong, L. Zhang, X. Wang, Y. Liu, and S. Zhou. 2021. A traditional Chinese medicine plant extract prevents alcohol-induced osteopenia. *Front. Pharmacol.* 12:754088.
- Qiao, Y., C. Liu, Y. Guo, W. Zhang, W. Guo, K. Oleksandr, and Z. Wang. 2022. Polysaccharides derived from *Astragalus membranaceus* and *Glycyrrhiza uralensis* improve growth performance of broilers by enhancing intestinal health and modulating gut microbiota. *Poult. Sci.* 101:101905.
- Qiu, Y., C. Tang, M. Serrano-Sosa, J. Hu, J. Zhu, G. Tang, C. Huang, and M. Huang. 2019. Bone microarchitectural parameters can detect oxytocin induced changes prior to bone density on mitigating bone deterioration in rabbit osteoporosis model using micro-CT. *BMC Musculoskelet. Disord.* 20:560.
- Ran, J., C. Ma, K. Xu, L. Xu, Y. He, S. A. A. Moqbel, P. Hu, L. Jiang, W. Chen, J. Bao, Y. Xiong, and L. Wu. 2018. Schisandrin B ameliorated chondrocytes inflammation and osteoarthritis via suppression of NF- κ B and MAPK signal pathways. *Drug Des. Dev. Ther.* 12:1195–1204.
- San Martin, R., R. Pathak, A. Jain, S. Y. Jung, S. G. Hilsenbeck, M. C. Pina-Barba, A. G. Sikora, K. J. Pienta, and D. R. Rowley. 2017. Tenascin-C and integrin α 9 mediate interactions of prostate cancer with the bone microenvironment. *Cancer Res.* 77:5977–5988.
- Santos, M. N., T. M. Widowski, E. G. Kiarie, M. T. Guerin, A. M. Edwards, and S. Torrey. 2022. In pursuit of a better broiler: tibial morphology, breaking strength, and ash content in conventional and slower-growing strains of broiler chickens. *Poult. Sci.* 101:101755.
- Settle, S. H. Jr., R. B. Rountree, A. Sinha, A. Thacker, K. Higgins, and D. M. Kingsley. 2003. Multiple joint and skeletal patterning defects caused by single and double mutations in the mouse *Gdf6* and *Gdf5* genes. *Dev. Biol.* 254:116–130.
- Shi, Z., L. Wang, J. Luan, L. Yin, X. Ji, W. Zhang, B. Xu, L. Chen, Y. He, R. Wang, and L. Liu. 2022. Exercise promotes bone marrow microenvironment by inhibiting adipin in diet-induced male obese mice. *Nutrients* 15:19.
- Spatier, D., T. P. Hill, J. O'Sullivan, R. M. Gruber, D. A. Conner, and C. Hartmann. 2006. Wnt9a signaling is required for joint integrity and regulation of *Ihh* during chondrogenesis. *Development* 133:3039–3049.
- Sun, P., Y. Zhang, Z. Wei, Z. Wang, S. Guo, and Y. Lin. 2020. Effect of Qing's decoction on leptin/leptin receptor and bone metabolism in naturally aging rats. *Evid. Based Complement. Altern. Med.* 2020:2532081.
- Suttapreyasri, S., S. Koontongkaew, A. Phongdara, and U. Leggat. 2006. Expression of bone morphogenetic proteins in normal human intramembranous and endochondral bones. *Int. J. Oral Maxillofac. Surg.* 35:444–452.
- Talbot, J. C., J. T. Nichols, Y. L. Yan, I. F. Leonard, R. A. BreMiller, S. L. Amacher, J. H. Postlethwait, and C. B. Kimmel. 2016. Pharyngeal morphogenesis requires *frs1-itga8*-dependent epithelial-mesenchymal interaction. *Dev. Biol.* 416:136–148.
- Tang, H., Y. Guo, Z. Zhang, Z. Li, Y. Zhang, Y. Li, X. Kang, and R. Han. 2020a. Integrative analysis of long non-coding RNA and mRNA in broilers with valgus-varus deformity. *PLoS One* 15:e0239450.

- Tang, X. F., Z. T. Ma, Y. Y. Gao, H. Wang, X. X. Li, P. Yu, and R. H. Liu. 2020b. Systemic osteoprotective effects of Epimedium folium and Ligustri lucidi Fructus in senile osteoporosis rats by promoting the osteoblastogenesis and osteoclastogenesis based on MLP-ANN model. *Chin. Med.* 15:87.
- Tang, D. Z., F. Yang, Z. Yang, J. Huang, Q. Shi, D. Chen, and Y. J. Wang. 2011. Psoralen stimulates osteoblast differentiation through activation of BMP signaling. *Biochem. Biophys. Res. Commun.* 405:256–261.
- Tian, W. X., J. K. Li, P. Qin, R. Wang, G. B. Ning, J. G. Qiao, H. Q. Li, D. R. Bi, S. Y. Pan, and D. Z. Guo. 2013. Screening of differentially expressed genes in the growth plate of broiler chickens with tibial dyschondroplasia by microarray analysis. *BMC. Genom.* 14:276.
- Twine, N. A., L. Harkness, J. Adjaye, A. Aldahmash, M. R. Wilkins, and M. Kassem. 2018. Molecular phenotyping of telomerized human bone marrow skeletal stem cells reveals a genetic program of enhanced proliferation and maintenance of differentiation responses. *JBM R Plus* 2:257–267.
- Wang, W., Y. Liu, J. Hao, S. Zheng, Y. Wen, X. Xiao, A. He, Q. Fan, F. Zhang, and R. Liu. 2016. Comparative analysis of gene expression profiles of hip articular cartilage between non-traumatic necrosis and osteoarthritis. *Gene* 591:43–47.
- Wang, F., H. Qian, L. Kong, W. Wang, X. Wang, Z. Xu, Y. Chai, J. Xu, and Q. Kang. 2021. Accelerated bone regeneration by astragaloside IV through stimulating the coupling of osteogenesis and angiogenesis. *Int. J. Biol. Sci.* 17:1821–1836.
- Waqas, M., H. Qamar, J. Zhang, W. Yao, A. Li, Y. Wang, M. Iqbal, K. Mehmood, X. Jiang, and J. Li. 2020. Puerarin enhance vascular proliferation and halt apoptosis in thiram-induced avian tibial dyschondroplasia by regulating HIF-1 α , TIMP-3 and BCL-2 expressions. *Ecotoxicol. Environ. Saf.* 190:110126.
- Waqas, M., Y. Wang, A. Li, H. Qamar, W. Yao, X. Tong, J. Zhang, M. Iqbal, K. Mehmood, and J. Li. 2019. Osthole: a coumarin derivative assuage thiram-induced tibial dyschondroplasia by regulating BMP-2 and RUNX-2 expressions in chickens. *Antioxidants (Basel)* 8:330.
- White, K. E., J. M. Cabral, S. I. Davis, T. Fishburn, W. E. Evans, S. Ichikawa, J. Fields, X. Yu, N. J. Shaw, N. J. McLellan, C. McKeown, D. Fitzpatrick, K. Yu, D. M. Ornitz, and M. J. Econs. 2005. Mutations that cause osteoglyphonic dysplasia define novel roles for FGFR1 in bone elongation. *Am. J. Hum. Genet.* 76:361–367.
- Williams, B., D. Waddington, D. H. Murray, and C. Farquharson. 2004. Bone strength during growth: influence of growth rate on cortical porosity and mineralization. *Calcif. Tissue Int.* 74:236–245.
- Winslow, M. M., M. Pan, M. Starbuck, E. M. Gallo, L. Deng, G. Karsenty, and G. R. Crabtree. 2006. Calcineurin/NFAT signaling in osteoblasts regulates bone mass. *Dev. Cell* 10:771–782.
- Witte, F., J. Dokas, F. Neuendorf, S. Mundlos, and S. Stricker. 2009. Comprehensive expression analysis of all Wnt genes and their major secreted antagonists during mouse limb development and cartilage differentiation. *Gene Exp. Patterns* 9:215–223.
- Wongin, S., R. Narkbunnam, S. Waikakul, P. Chotiyarnwong, T. Aresanasuwan, S. Roytrakul, and K. Viravaidya-Pasuwat. 2021. Construction and evaluation of osteochondral-like tissue using chondrocyte sheet and cancellous bone. *Tissue Eng. Part A* 27:282–295.
- Wu, H., Z. Wu, P. Li, Q. Cong, R. Chen, W. Xu, S. Biswas, H. Liu, X. Xia, S. Li, W. Hu, Z. Zhang, S. L. Habib, L. Zhang, J. Zou, H. Zhang, W. Zhang, and B. Li. 2017. Bone size and quality regulation: concerted actions of mTOR in mesenchymal stromal cells and osteoclasts. *Stem Cell Rep.* 8:1600–1616.
- Wu, T., Z. Yao, G. Tao, F. Lou, H. Tang, Y. Gao, X. Yang, and J. Xiao. 2021. Role of Fzd6 in regulating the osteogenic differentiation of adipose-derived stem cells in osteoporotic mice. *Stem Cell Rev. Rep.* 17:1889–1904.
- Xiao, Y., and X. P. Zhao. 2019. Screening pathways and hub genes involved in osteoclastogenesis by gene expression analysis of circulating monocytes based on Gibbs sampling. *Exp. Ther. Med.* 17:2529–2534.
- Xiao, L., M. Zhong, Y. Huang, J. Zhu, W. Tang, D. Li, J. Shi, A. Lu, H. Yang, D. Geng, H. Li, and Z. Wang. 2020. Puerarin alleviates osteoporosis in the ovariectomy-induced mice by suppressing osteoclastogenesis via inhibition of TRAF6/ROS-dependent MAPK/NF-kappaB signaling pathways. *Aging (Albany NY)* 12:21706–21729.
- Xiong, Y., R. Sun, J. Li, Y. Wu, and J. Zhang. 2020. Latent TGF-beta binding protein-1 plays an important role in craniofacial development. *J. Appl. Oral Sci.* 28:e20200262.
- Xu, T. T., P. Chen, C. D. Zhang, A. Shaukat, L. X. Lin, K. Yue, W. L. Ding, X. Tong, K. L. Liu, Y. F. He, J. F. Xie, F. Liu, C. Zhang, H. Y. Zhang, and S. C. Huang. 2023. Gut microbiome dysregulation drives bone damage in broiler tibial dyschondroplasia by disrupting glucose homeostasis. *NPJ Biofilms Microbiom.* 9:1.
- Xu, T., K. Yue, C. Zhang, X. Tong, L. Lin, Q. Cao, and S. Huang. 2022a. Probiotics treatment of leg diseases in broiler chickens: a review. *Probiotics Antimicrob. Proteins* 14:415–425.
- Xu, T., J. Zheng, W. Jin, L. Li, L. Lin, A. Shaukat, C. Zhang, Q. Cao, M. Ashraf, and S. Huang. 2022b. Total flavonoids of rhizoma drynariae ameliorate bone growth in experimentally induced tibial dyschondroplasia in chickens via regulation of OPG/RANKL axis. *Front. Pharmacol.* 13:881057.
- Yan, C. P., X. K. Wang, K. Jiang, C. Yin, C. Xiang, Y. Wang, C. Pu, L. Chen, and Y. L. Li. 2022. β -Ecdysterone enhanced bone regeneration through the BMP-2/SMAD/RUNX2/Osterix signaling pathway. *Front. Cell. Dev. Biol.* 10:883228.
- Yang, W., M. He, J. Zhao, and Z. Wang. 2014. Association of ITGA3 gene polymorphisms with susceptibility and clinicopathological characteristics of osteosarcoma. *Med. Oncol.* 31:826.
- Yang, X., Y. X. Mai, L. Wei, L. Y. Peng, F. X. Pang, L. J. Wang, Z. P. Li, J. F. Zhang, and A. M. Jin. 2023. MLK3 silence suppressed osteogenic differentiation and delayed bone formation via influencing the bone metabolism and disturbing MAPK signaling. *J. Orthop. Translat.* 38:98–105.
- Yang, J., L. Qin, J. Huang, Y. Li, S. Xu, H. Wang, S. Zhu, J. Wang, B. Zhu, F. Li, W. Huang, X. Gong, and N. Hu. 2021. Astragalus polysaccharide attenuates LPS-related inflammatory osteolysis by suppressing osteoclastogenesis by reducing the MAPK signalling pathway. *J. Cell. Mol. Med.* 25:6800–6814.
- Yerges, L. M., Y. Zhang, J. A. Cauley, C. M. Kammerer, C. S. Nestlerode, V. W. Wheeler, A. L. Patrick, C. H. Bunker, S. P. Moffett, R. E. Ferrell, and J. M. Zmuda. 2009. Functional characterization of genetic variation in the Frizzled 1 (FZD1) promoter and association with bone phenotypes: more to the LRP5 story? *J. Bone Miner. Res.* 24:87–96.
- Yoon, Y. M., C. D. Oh, D. Y. Kim, Y. S. Lee, J. W. Park, T. L. Huh, S. S. Kang, and J. S. Chun. 2000. Epidermal growth factor negatively regulates chondrogenesis of mesenchymal cells by modulating the protein kinase C-alpha, Erk-1, and p38 MAPK signaling pathways. *J. Biol. Chem.* 275:12353–12359.
- Yu, D., Z. Li, J. Cao, F. Shen, and G. Wei. 2022. microRNA-25-3p suppresses osteogenic differentiation of BMSCs in patients with osteoporosis by targeting ITGB3. *Acta Histochem.* 124:151926.
- Zhang, X. Y., Y. P. Chen, C. Zhang, X. Zhang, T. Xia, J. Han, N. Yang, S. L. Song, and C. H. Xu. 2020b. Icariin accelerates fracture healing via activation of the WNT1/beta-catenin osteogenic signaling pathway. *Curr. Pharm. Biotechnol.* 21:1645–1653.
- Zhang, H., S. Li, J. Lu, J. Jin, G. Zhu, L. Wang, Y. Yan, L. He, B. Wang, X. Wang, and H. Yu. 2021. α -Cyperone (CYP) down-regulates NF- κ B and MAPKs signaling, attenuating inflammation and extracellular matrix degradation in chondrocytes, to ameliorate osteoarthritis in mice. *Aging (Albany NY)* 13:17690–17706.
- Zhang, H., Y. Wang, K. Mehmood, Y. F. Chang, Z. Tang, and Y. Li. 2020a. Treatment of tibial dyschondroplasia with traditional Chinese medicines: "lesson and future directions". *Poult. Sci.* 99:6422–6433.
- Zhang, Z., L. Xiang, D. Bai, X. Fu, W. Wang, Y. Li, H. Liu, J. Pan, Y. Li, G. G. Xiao, and D. Ju. 2014. Treatment with Rhizoma dioscoreae extract has protective effect on osteopenia in ovariectomized rats. *Sci. World J.* 2014:645975.
- Zhang, J., H. Yu, H. Zhang, Q. Zhao, W. Si, Y. Qin, and J. Zhang. 2023. Dietary Epimedium extract supplementation improves intestinal functions and alters gut microbiota in broilers. *J. Anim. Sci. Biotechnol.* 14:14.
- Zheng, Y., Q. Wang, T. Li, J. Qian, Y. Lu, Y. Li, E. Bi, F. Reu, Y. Qin, J. Drazba, E. Hsi, J. Yang, Z. Cai, and Q. Yi. 2016. Role of myeloma-derived MIF in myeloma cell adhesion to bone

- marrow and chemotherapy response. *J. Natl. Cancer Inst.* 108: djw131.
- Zhou, Z. Q., S. Ota, C. Deng, H. Akiyama, and P. J. Hurlin. 2015. Mutant activated FGFR3 impairs endochondral bone growth by preventing SOX9 downregulation in differentiating chondrocytes. *Hum. Mol. Genet.* 24:1764–1773.
- Zhou, Z., and W. C. Xiong. 2011. RAGE and its ligands in bone metabolism. *Front. Biosci. (Schol. Ed.)* 3:768–776.
- Zimmer, S. N., Q. Zhou, T. Zhou, Z. Cheng, S. L. Abboud-Werner, D. Horn, M. Lecoche, R. White, A. V. Krivtsov, S. A. Armstrong, A. L. Kung, D. M. Livingston, and V. I. Rebel. 2011. Crebbp haploinsufficiency in mice alters the bone marrow microenvironment, leading to loss of stem cells and excessive myelopoiesis. *Blood* 118:69–79.
- Zuidhof, M. J., B. L. Schneider, V. L. Carney, D. R. Korver, and F. E. Robinson. 2014. Growth, efficiency, and yield of commercial broilers from 1957, 1978, and 2005. *Poult. Sci.* 93 2970-29.